MILITARY SPECIFICATION

FORWARD LOOKING RADAR SET, AN/APQ-126(V)

This specification has been approved by the Naval Air Systems Command Department of the Navy.

1. SCOPE

1.1 SCOPE - This specification covers the design, performance, and test requirements for a forward looking radar set which shall provide the following functions:

Terrain following computations (TF)

Terrain avoidance (TA)

Ground mapping (shaped beam) (GMS)

Ground mapping (pencil beam) (GMP)

Air-to-ground ranging (AGR)

Beacon interrogate and receive (BCN)

Walleye television display (TV Display)

Shrike Improved Display System (SIDS)

Radar navigation updating (CMPTR)

Radar bombing (RDR BOMB)

These functions, as provided, shall be compatible with the flight and control characteristics of the A-7D/E airplane, a high performance light attack airplane with all weather navigation, target acquisition, and weapon delivery capabilities.

These additional functions are available when the radar set is configured with an Indicator IP-1256/APQ-126 (V) and Convertor CV-3368/APQ-126 (V):

Forward Looking Infrared (FLIR) display (when the necessary input signals are provided).

Comments or recommendations which may be of use in improving this document may be sent to the preparing activity when a self-addressed standardization document improvement proposal (DD Form 1426) appears at the end of this document, or by letter addressed to the cognizant activity.

FSC 5841

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Expanded map display (EXPAND) (when used in conjunction with an appropriately modified Power Supply Programmer PP-6130/APQ-126 (V)).

Frozen map display (HOLD).

1.2 Classification - The Radar Set, AN/APQ-126(V), shall consist of the following items:

Item	Type Designation	Applicable Paragraph
Power Supply - Programmer, Antenna Scan (P/O Fwd Ass'y Grp)	PP-6130/APQ-126(V)	3.5.1
Transmitter, Radar (P/O Fwd Ass'y Grp)	T-1091/APQ-126(V)	3.5.2
Antenna-Receiver (P/O Fwd Ass'y Grp)	AS-2272/APQ-126(V)	3.5.3
Computer, Air Navigation (P/O Fwd Ass'y Grp)	CP-954/APQ-126(V)	3.5.4
Control, Radar Set (Red Lighting)	C-8251/APQ-126(V)	3.5.5
OR		
Control, Radar Set (White Lighting) (P/O Cockpit Grp)	C-8252/APQ-126(V)	3.5.5
Generator, Sweep (P/O Cockpit Grp)	SG-811/APQ-126(V)	3.5.7
OR		
Convertor, Signal Data (P/O Cockpit Grp)	CV-3386/APQ-126(V)	3.5.12
Fault Locator, Radar Set (P/O Cockpit Grp)	MX-8195/APQ-126(V)	3.5.8
Indicator, Multiple, Air Navigation (Red Lighting)	IP-940/APQ-126(V)	3.5.9
OR		
Indicator, Multiple, Air Navigation (White Lighting) (P/O Cockpit Grp)	IP-952/APQ-126(V)	3.5.9
OR		
Indicator, Intra Target Data, FWD (Red Lighting) (P/O Cockpit Grp)	IP-1256/APQ-126(V)	3.5.13

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Item	Type Designation	Applicable Paragraph
Control, Radar Set (Red Lighting) OR Control, Radar Set (Instrument Panel Mount-White Lighting) (P/O Cockpit Grp)	C-8253/APQ-126(V) OR C-8255/APQ-126(V)	3.5.6
Base, Shock Mount, Electrical Equipment (P/O Fwd Ass'y Grp)	MT-4043/APQ-126(V)	3.5.10
Fan, Centrifugal (P/O Fwd Ass'y Grp)	HD-821A	3.5.11

1.3 Associated Equipment - This radar set shall operate with the associated equipment listed in 6.6.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents form a part of this specification to the extent specified herein. Listed are the issues of the documents in effect for the previous procurement. However, in lieu of the issue listed, the contractor shall use the latest issue in effect of these documents where feasible. If the use of the latest issue will affect design performance, or interchangeability of any replaceable part, then the issue of the document listed below shall be used.

SPECIFICATIONS

Military

MIL-T-27	Transformers and Inductors (Audio, Power, and High Power Pulse); General Specification for
MIL-W-5088B-1	Wiring Aircraft, General Specification for
MIL-E-5400F(ASG)-1	Electronics Equipment, Aircraft, General Specification for
MIL-T-5422E(ASG)-2	Testing, Environmental, Aircraft Electronic Equipment

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MIL-I-6181D	Interference Control Requirements, Aeronautical Equipment
MIL-C-6781B(ASG)	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-M-7793C-1	Meter, Time Totalizing, 380-420 Hz, General Specification for
MIL-D-18300B-1	Design Data Requirements for Contracts Covering Airborne Electronic Equipment
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-E-19600A	Electronic Modules, General Aircraft Requirements for
MIL-T-21200D(ASG)-1	Test Equipment for use with Electronic and Fire-Control Systems, General Specification for
MIL-I-23524	Indicator, Altitude-Director, Remote
MIL-A-23887	Altimeter, Radar AN/APN-141
MIL-C-38999	Connectors, Electrical, Miniature, Quick Disconnect with Removable Crimp Type Contacts (for Special Weapons System Circuitry), Established Reliability

Other

AS-1720	Convertor, Signal Conditioner
WS3180A	Detail Specification, Guidance Section, Guided Weapon, MK 27 Mod 0
XWS-6517	Convertor, Signal Conditioner

STANDARDS

Military

MIL-STD-129C	Marking for Shipment and Storage

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MIL-STD-143A	Specifications and Standards, Order of Precedence for the Selection of Equipment
MIL-STD-210A	Climatic Extremes for Military Equipment
MIL-STD-449B	Radio Frequency Spectrum Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756A	Reliability Prediction
MIL-STD-781A	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for System and Equipment Development and Production
MIL-STD-794	Parts and Equipment Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment

Electronic Industries Association

EIA-RS-343A	High Resolution Monochrome Closed Circuit Television Camera, Electrical Performance Standards for

Vought Systems Division

204-20-21A	Digital Scan Converter for TA-7C Airplanes, Procurement Specification for

PUBLICATIONS

Military

MIL-HDBK-217A	Reliability Stress and Failure Rate Data for Electrical Equipment

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2.1.1 Precedence of Specifications - Where the requirements of this procurement specification and those of any other applicable specifications differ, the requirements of this procurement specification shall govern. In all other instances, publications shall be used in the order of precedence established by MIL-STD-143.

2.1.2 Deviations - Deviations to this procurement specification or any specification applicable to the equipment procured hereunder are not authorized without prior written approval of NAVAIR.

2.1.3 Availability of Documents - When requesting specifications, standards, and publications, refer to both title, and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

2.1.3.1 Availability of Electronic Industries Association Documents - Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, N.W., Washington, DC 20006.

2.1.3.2 Availability of Vought Systems Division Documents - Application for copies should be addressed to the Vought Systems Division, LTV Aerospace Corporation, P. O. Box 5907, Dallas, Texas 75222.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to Specification MIL-E-5400 for Class 2, Category III equipment.

c. Nonrepairable subassemblies, as outlined in Specification MIL-E-5400 for Class 2, Category III equipment, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies shall be reliable.

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d. When previously produced models of this radar set did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors, and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400 for Class 2, Category III equipment.

3.2.2 Microelectronic Modular Assemblies - Not applicable.

3.2.3 Modules - The use of modular space assignment, and the construction of plug-in subassemblies is encouraged in the design of equipment specified herein. Small groups of subassemblies may constitute a larger subassembly where necessity dictates the use of common pressurization, cooling or shielding. The use of potted or sealed repairable subassemblies which causes difficulty in part replacement shall require prior approval. Subassemblies shall be designed in accordance with MIL-E-19600 excluding the dimension requirements. The subassembly dimensions and power supply requirements shall be as best suited for the application. Subassemblies shall be keyed to prevent inadvertent misinstallation.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for Class 2, Category III equipment for design, construction, and workmanship except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the radar set, excluding cables, shall be a minimum consistent with good design and shall not exceed 239.5 pounds (108.6 kg).

3.3.2 Reliability - The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.3.2.1 Operational Stability - The radar set shall operate with satisfactory performance, continuously, or intermittently for a period of at least 100 hours when operated in a temperature environment not exceeding that specified in paragraph 3.3.10.2 without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The radar set shall have a minimum total operating life of 4000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

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3.3.2.3 Storage - The equipment shall be designed to operate satisfactorily after five years storage and transit in accordance with MIL-STD-210.

3.3.2.4 Reliability in Mean Time Between Failure (MTBF) - The equipment shall demonstrate a specified MTBF of 106 hours when tested in accordance with paragraph 4.4 herein. The equipment reliability shall not be degraded below the required level as a result of normal field usage and maintenance.

3.3.2.5 Time Totalizing Meter - The following units shall contain time totalizing meters in accordance with Specification MIL-M-7793.

Unit	Meter
Antenna - Receiver	MS17322-6A
Power Supply-Programmer, Antenna Scan	MS17322-6A
Computer, Air Navigation	MS17322-6A
Transmitter, Radar	MS17322-4
Indicator, Multiple, Air Navigation	MS17322-6A
Sweep Generator	MS17322-6A
Indicator, Intra Target Data, FWD	MS-17322-10
Convertor, Signal Data	MS-17322-10

3.3.3 Cabling and Connections -

3.3.3.1 Connectors - Except for those connectors used on the forward radar assembly, all connectors of the radar system used for testing purposes or interconnection between units of the radar or other aircraft systems shall be in accordance with MIL-C-38999 or as otherwise approved by the procuring activity.

3.3.3.2 Interconnection Cabling - The radar set shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

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3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781, Type 1. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability - The radar set shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.6 Interference Control - The generation of radio interference by the radar set and the vulnerability of the radar set to radio interference shall meet the requirements of Specification MIL-I-6181.

3.3.7 Provisions for Maintainability - The basic radar set design shall comply with all applicable portions of MIL-STD-470. The equipment shall be designed so that standard tools and test equipment may be used to the greatest possible extent to accomplish all necessary adjustments, maintenance, and testing. A proposed list and required information for standard and special tools and test equipment shall be submitted to NAVAIR for approval.

3.3.7.1 Maintainability Requirement - Field (intermediate) level maintenance shall not exceed 3.0 direct man-hours mean time to repair (MTTR) per malfunction. No more than 12.0 man-hours of direct time shall be required to repair any single failure. Preventive maintenance shall require no more than 6.0 man-hours direct time to accomplish and shall not be required more often than an interval of 500 flight hours. Repair times shall be computed on the basis of accomplishment by maintenance personnel equivalent to Air Force skill level 5. Information on the number of personnel required for each of the above tasks will be provided. All organizational level maintenance must be capable of being performed under severe operational meteorological environments using only standard hand tools.

3.3.7.1.1 Design objectives, for maintainability, shall permit all maintenance actions to be accomplished at organizational or intermediate level. Line replaceable units shall be replaceable at the organizational level using procuring activity approved harmonization techniques.

3.3.7.1.2 Modular type construction shall be used so that faulty circuits can be readily replaced. Non-repairable subassemblies may be used, and are preferred, providing such subassemblies cost less than $75.00, except where significant weight savings may be achieved or where mechanical integration with or modification of existing design requires encapsulated subassemblies or in the case of microwave

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energy devices. All nonrepairable subassemblies require prior approval of the procuring activity maintainability before release for production. All replaceable subassemblies should be capable of replacement without use of solder or weld. Any subassembly designed to be soldered or welded into its circuit will be submitted to the procuring activity for approval prior to production incorporation.

3.3.7.1.3 Arrangement for each line replaceable unit (LRU) shall make provisions for rapid removal and replacement of all replaceable items without removal of other parts, subassemblies, etc.

3.3.7.1.4 Test points shall be external and all internal check points for troubleshooting shall be accessible without removing other subassemblies, covers, etc. All calibration adjustments shall be accessible without need to remove items or open circuits. Requirements for jumpers in the system test receptacles should be avoided. In the event jumpers are required, provisions shall be included to indicate a system malfunction when the system is turned on with the cap removed.

3.3.7.1.5 Unit packaging requires approval of the procuring activity prior to release. Drawing submittal shall contain sufficient detail to evaluate use of internal connectors, plugs, terminals, etc.

3.3.7.1.6 The equipment shall be designed to include the following:

The equipment shall be designed for compatibility with automatic checkout equipment and shall include test provisions to permit diagnostic and performance verification testing at intermediate and depot level maintenance.

Each LRU shall have sufficient accessible external test points that, when used in conjunction with the operating connector will

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(1) permit quantitative determination of the performance level of the LRU, (2) if substandard performance is determined, permit isolation of fault to the removed subassembly. Subassembly refers to the first level of mechanical disassembly (module, subassembly, etc.) within the LRU.

Each subassembly within the LRU shall have exposed or accessible internal test points when disassembled to the extent necessary for intermediate and depot level maintenance. These internal test points shall be sufficient to fulfill the requirements of external test points (above), except for the level below the lowest repairable assembly. That is, a packaging unit (module, board, PC card, etc.) that is designated Not Repairable need not meet the requirements for fault isolation.

3.3.7.2 Built-in Test Equipment - Built-in test equipment construction and packaging, and provisions for test points chall be as specified in all applicable portions of Specification MIL-STD-470. The radar set shall include built-in test provisions for preflight and troubleshooting purposes. The self-test feature shall be capable of isolating 80% of all equipment malfunctions to the faulty LRU, and as a design objective shall function independently of other system inputs. Provisions shall be made to verify AGR operation and the Radar/Ranging/Computer interface.

3.3.8 Nomenclature, Nameplates, and Identification Marking - Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (25°C ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	115 ± 1.0 volt 400 Hz and 27.5 ± 0.5 volt dc

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3.3.10 Service Conditions - The radar set shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.3.10.1 Vibration - The radar set shall operate satisfactorily when subjected to the vibration requirements of the following curves:

3.3.10.1.1 Radar Forward Assembly and Centrifugal Fan with Divergent Duct - Figure 1 of this specification shall apply.

3.3.10.1.2 Sweep Generator and Signal Data Convertor - Figure 2 of this specification shall apply.

3.3.10.1.3 Fault Locator and Radar Set Controls - Figure 3 of this specification shall apply.

3.3.10.1.4 Indicator - Figure 4 of this specification shall apply.

3.3.10.2 Temperature/Altitude - The radar set shall operate as follows:

3.3.10.2.1 The Forward Assembly Group, Indicator, and Radar Set Control shall meet the requirements of MIL-E-5400 for Curve A of Class 1 equipment.

3.3.10.2.2 The items of the radar set not covered by 3.3.10.2.1 shall meet the requirements of MIL-E-5400 for Curve A of Class 2 equipment except the maximum altitude shall be reduced to 50,000 feet (15.24 km).

3.3.10.3 Altitude - See paragraphs 3.3.10.2.1 and 3.3.10.2.2.

3.3.10.4 Shock - The equipment shall be capable of withstanding shock load as follows:

3.3.10.4.1 Vibration isolators in place: 15 g's and a time duration of 11 $\pm$ 1 milliseconds.

3.3.10.4.2 Mounting base (crash safety): 30 g's.

3.3.10.4.3 The forward assembly, fault isolator, and radar set control shall also be capable of withstanding the following shock loads:

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Peak Acceleration	Time-to-Peak Acceleration
15 g's	0.0025
21 g's	0.00125
27 g's	0.00083
31.5 g's	0.000675
45 g's	0.0005
60 g's	0.0004

The time-duration-to-peak acceleration refers to the 0.25 cycle time period of a sinusoidal waveform.

3.3.10.5 Humidity - 100 percent at temperature to +55°C, applicable to all units.

3.3.10.6 Sand and Dust - The radar set shall withstand, in both operating and nonoperating condition, exposure to sand and dust particles as encountered in desert areas.

3.3.10.7 Fungus - The radar set shall withstand, in both the operating and nonoperating condition, exposure to fungus growth as encountered in tropical climates.

3.3.10.8 Salt Atmosphere - The radar set shall withstand, in both the operating and nonoperating condition, exposure to a salt-sea atmosphere.

3.3.10.9 Explosive Conditions - The radar set shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.3.10.10 Acoustic Noise - The radar set shall be capable of satisfactory operation under the following acoustic environment.

3.3.10.10.1 Cockpit Mounted Equipment - An overall noise level of 120 ± 5 dB having a relative pressure distribution in accordance with Figure 515-1 of MIL-STD-810.

3.3.10.10.2 Internally Mounted Equipment - (not in cockpit) - An overall noise level of 135 ± 5 dB having a relative pressure distribution in accordance with Figure 515-1 of MIL-STD-810.

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3.3.10.11 Maneuvering Load - The radar set shall withstand and operate satisfactorily under maneuvering acceleration as follows:

Forward Radar Assembly:

3.3.10.11.1 Vertical -4 to +8.5 g's.

3.3.10.11.2 Catapulting 6.0 g's.

3.3.10.11.3 Arresting 6.0 g's.

3.3.10.11.4 Lateral ± 2.5 g's.

Other Components:

3.3.10.11.5 Vertical -3 to +8.5 g's.

3.3.10.11.6 Catapulting 6.0 g's.

3.3.10.11.7 Arresting 6.0 g's.

3.3.10.11.8 Lateral ± 2.5 g's.

3.3.11 Warm-Up Time - The time required for the radar set to warm up prior to operation shall be kept to a minimum and shall not exceed 200 seconds.

3.3.12 Input Electrical Power -

3.3.12.1 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

a. AC Power (Single Phase), not applicable.

b. AC Power (Three Phase) 115/200 V Category B, 2500 VA

c. DC Power 28 V, Category B, 225 watts

3.3.12.2 Lighting Power - Input power for lighting shall require not more than 2 amps at 5 volts 400 Hz.

3.3.12.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the radar set.

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3.3.12.4 Standby Provisions - The equipment shall provide for a Standby position wherein complete operation shall be available except for the transmitter and associated circuitry and antenna scan operation.

3.3.13 Cooling - The radar equipment shall be capable of being cooled by natural convection and radiation, if practical. An exhaust or recirculation fan (or fans) mounted intergral with each box may be used if necessary. No conditioned air will be required or supplied except as noted in paragraph 3.3.13.1. The radar shall be protected from component failure in the event of loss of cooling air. This shall be accomplished by inclusion of a thermostatically controlled switch located in the forward assembly.

3.3.13.1 Cooling Air Provisions - Cabin exhaust air will be supplied through a shroud which covers but is not connected to the cooling air inlet opening on the forward assembly as specified in paragraph 3.3.13.1.1. Conditioned air will be supplied through a shroud which covers but is not connected to the cooling air inlet opening on the sweep generator (or Convertor CV-3368/APQ-126(V)) as specified in paragraph 3.3.13.1.2.

3.3.13.1.1 Forward Assembly Cooling Air Distribution Provisions - Cabin exhaust air will be supplied to the cooling air inlet opening in the forward-assembly through a shroud which covers but does not connect to the equipment. The cabin exhaust air will normally be supplied at a temperature of less than 80°F. Minimum flow rate shall be no less than 7.5 pounds (3.402 kg) per minute at sea level, decreasing to 6.5 pounds (2.948 kg) per minute at 8000 feet (2.438 km) and 4.25 pounds (1.928 kg) per minute at 50,000 feet (15.24 km). A fan shall be provided by the contractor, integral with the equipment, to circulate the cooling air through the forward assembly at all times. When the cabin exhaust air flow is interrupted or turned off the radome ambient air shall be circulated through the equipment by the fan. The radome ambient in flight is considered to be a MIL-E-5400 Class 1 Thermal Environment.

3.3.13.1.2 Sweep Generator Cooling Air Distribution Provisions - Conditioned air will be supplied through a shroud which is around, but not connected to, the cooling air inlet on the sweep generator (or Convertor CV-3368/APQ-126(V)) to maintain a MIL-E-5400 Class 1 Thermal Environment with a cooling capacity of 300 watts. A fan (or fans) shall be provided by the contractor, integral with the equipment, to circulate the conditioned air through the sweep generator (or Convertor CV-3368/APQ-126(V)) as required. With the air conditioner inoperative, compartment ambient air would be circulated through the equipment by the fan. The compartment ambient in flight iswithin the MIL-E-5400 Class II thermal limit. During the emergency operation, i.e., when the air conditioning system is inoperative, the equipment shall continue to operate satisfactorily up to a maximum compartment temperature equivalent to MIL-E-5400 Class II.

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3.3.14 Pressurization Air Provisions - The forward assembly shall contain a pressure line for connecting aircraft supplied pressurization air to the waveguide. The pressurization air supplied shall meet the following requirements:

a. Pressure: 15 to 19 psia (1.055 to 1.336 kg/cm^2) (normal) 24 psia (1.758 kg/cm^2) max. (relief valve operation provided for regulator failure condition)

b. Flow: 20 standard cubic inches (327 standard cm^3) per minute, maximum

c. Relative Humidity: 20% (Maximum at 70°F)

3.3.15 Oil Filled Components - Any design which employs oil filled parts shall be such that no explosive hazard exists.

3.3.16 Spectrum Signature - Spectrum signature data in accordance with MIL-STD-449 shall be provided to the extent applicable to radar, tracking systems, monopulse, amplitude comparisons, for the transmitter/antenna complex.

3.3.17 Servo Response - This equipment shall contain stabilization servo mechanisms that adhere to the following closed-loop, linear response characteristics.

Servo	Damping Ratio (min)	Natural Freq. (min)
Elevation	0.3	6.5 Hz
Azimuth	0.2	4.0 Hz
Roll	0.2	5.0 Hz

The equipment shall, in addition, respond to a 10 degree (0.175 rad) step input at a minimum rate of 126 degrees (2.199 rad) per second (elevation), 31.5 degrees (0.55 rad) per second (azimuth) and 31.5 degrees (0.55 rad) per second (roll).

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances and values setting forth acceptable variations from the performance under the standard conditions will be specified. Performance requirements for the CV-3386/APQ-126(V) Signal Data Converter and the IP-1256/APQ-126(V) Indicator are specified in Vought Systems Division Specification 204-20-21A.

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3.4.1 Operation - The radar shall provide for the following operational capabilities, selectable by the pilot, and subject to conditions in Table 1.

3.4.1.1 Ground mapping, shaped beam (GMS)

3.4.1.2 Ground mapping, pencil beam (GMP)

3.4.1.3 Terrain following (TF)

3.4.1.4 Terrain avoidance (TA)

3.4.1.5 Combined terrain following/terrain avoidance (CSTA)

3.4.1.6 Combined terrain following/ground mapping (CSGMP)

3.4.1.7 Air-to-ground ranging, clutter/isolated targets (AGR)

3.4.1.8 Beacon (BCN)

3.4.1.9 Television (TV) display

3.4.1.10 SIDS display

3.4.1.11 FLIR display (when configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V) and when the necessary inputs to the radar are provided).

3.4.1.12 Expanded Map display (when configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V) and an appropriately modified Power Supply PP-6130/APQ-126(V)).

3.4.1.13 Frozen Map display (HOLD) (when configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V)).

3.4.2 Mode Requirements - The required performance for each mode shall be as follows:

3.4.2.1 Ground Mapping (Shaped Beam) - The radar map display shall be a depressed center plan position indicator (PPI) type with azimuth scan coverage of ± 45 degrees (0.785 rad) centered about the ground track. Solid ground painting is required. The detection of land-water boundaries, large ships, cities, and other hard targets shall be possible at maximum radar range. The detection of tactical targets such as small industrial complexes, tank farms, airfields, highway intersections, bridges, etc. shall be possible at shorter ranges.

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3.4.2.1.1 Manual Range Cursor - During the mapping mode, a range cursor shall be provided consisting of a constant range line, manually adjustable in range position by the pilot, as controlled by the radar control specified in paragraph 3.4.22 and intensity as controlled by paragraph 3.5.9.4.8.

3.4.2.1.2 Map Ranges - Map ranges of 5, 10, 20, 40 and 80 sea miles shall be selectable by the pilot.

3.4.2.1.3 Antenna Function - In the ground mapping (shaped beam) mode, the antenna beam shall be cosecant^2 θ cosine θ function in elevation, whose characteristics are optimized for ground mapping at 40,000 feet (12.192 km) altitude and at a slant range between 33 and 36 sea miles. The antenna gain characteristics shall be as specified in paragraph 3.4.20.2.1.

3.4.2.1.4 Antenna Scan - The antenna shall be programmed in a roll, pitch, and drift stabilized single bar azimuth pattern with scan coverage limits and stabilization as specified in paragraph 3.4.20.4. Control shall be provided to manually tilt the antenna as defined in paragraph 3.4.2.7.2.

3.4.2.1.5 Indicator Scope Display - The display shall be presented on the radar indicator as a drift stabilized depressed center section PPI scan on a ground range scale. Display accuracy shall be as specified in paragraphs 3.4.3.1 and 3.4.3.2. Azimuth coverage shall be at least $\pm$ 45 degrees (0.785 rad) about the flight path.

3.4.2.1.6 Range and Azimuth Cursors - Upon receipt of a discrete signal as defined in paragraph 3.4.9.18 the radar shall display a range cursor and an azimuth cursor. The cursor display shall consist of a range line and an azimuth line on the indicator in accordance with range and azimuth information provided by the Navigation/Weapon Delivery Computer as defined in paragraph 3.4.9.19. Intensity of the cursor display shall be controlled as defined in paragraph 3.5.9.4.8.

3.4.2.2 Ground Mapping (Pencil) - The pencil beam ground mapping mode shall function the same as the shaped beam ground map mode as specified in paragraphs 3.4.2.1 through 3.4.2.1.6, except as follows:

3.4.2.2.1 The antenna beam shall be a pencil beam whose characteristics are specified in paragraph 3.4.20.2.2.

3.4.2.2.2 Weather return rejection, as specified in paragraph 3.4.5, shall be provided as an aid in improving the map display.

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3.4.2.3 Terrain Avoidance Mode (TA) - The terrain avoidance mode shall utilize a single bar azimuth scan and generate a depressed center, sector PPI-type presentation which displays that terrain which protrudes above the aircraft's altitude.

3.4.2.3.1 Antenna Beam - The antenna beam shall be an elevation monopulse as specified in paragraph 3.4.20.2.2.

3.4.2.3.2 Antenna Scan - The antenna shall be programmed in a roll, pitch, and drift stabilized single bar azimuth pattern with coverage limits and stabilization as specified in paragraph 3.4.20.4.

3.4.2.3.3 Range Coverage - Ranges of 5 and 10 sea miles shall be provided for selection when operating in the TA mode. Provision shall be made to automatically limit the range to 10 sea miles if a greater range is selected.

3.4.2.3.4 Radar Indicator Scope Display - The radar indicator display shall be presented as a drift stabilized depressed center, sector PPI scan in coordinates of slant range and azimuth. The TA display shall present video representing only that terrain which protrudes above the aircraft's altitude. The display accuracy shall be as specified in paragraph 3.4.3.4.

3.4.2.3.5 Weather Rejection - The weather return rejection feature, as described in paragraph 3.4.5, shall be provided as an aid to improve the terrain avoidance display presentation during adverse atmospheric conditions.

3.4.2.4 Terrain Following Mode (TF) - The terrain following mode shall vertically scan the terrain along a plane determined by the ground track of the airplane (corrected for turns in accordance with paragraph 3.4.2.4.1.4). It shall present an E^2 scan type display on the radar indicator and provide an elevation flight command (climb/dive) signal suitable for driving a horizontal pointer on an Attitude Director Indicator (ADI) and Head-Up Display as defined in paragraph 3.4.2.4.6. Mechanization of the TF mode shall be in accordance with paragraphs 3.4.2.4.1 through 3.4.2.4.8.

3.4.2.4.1 Terrain Following Control Requirements - In the TF mode, the radar angle and range data plus command and video generation and display shall be sufficiently accurate to enable the aircraft to be flown safely at clearance altitude settings from 200 to 2000 feet (60.96 to 609.6 m). Discrete detented altitude settings of 200, 500, 1000, 1500, and 2000 feet (60.96, 152.4, 304.8, 457.2 and 609.6 m) shall be provided. The system shall provide proper TF control and rapid response consistent with good stability. Control information presented to the pilot shall be processed to provide well damped, accurate, flight-path control. Smooth

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operation is required with particular attention to limiting the rate of change of acceleration. Flight-path characteristics resulting from changes in clearance altitude setting shall be rapid but smooth. Switching transients resulting from mode selection shall be minimized.

3.4.2.4.1.1 Flight-Path Requirements - As a design objective, the aircraft installed system shall exhibit the following TF characteristics during simulation:

3.4.2.4.1.1.1 TF system control over flat terrain or water shall result in standard deviation of clearance $\sigma \leq 30$ feet (9.144 m) for straight line and turning flight with bank angles in a coordinated turn up to 10 degrees (0.175 rad) at mach 0.4 and 20 degrees (0.349 rad) at mach 0.8.

3.4.2.4.1.1.2 The undershoot resulting from an approach to flat terrain or water from maximum descent dive angle shall be less than 50 feet (15.24 m) below the set clearance. The maneuver shall be smoothly executed and settle out within $\pm$ 20 feet (6.096 m) of the set clearance in less than three times, the shortest time possible for the maneuver within the pull up "g" restrictions.

3.4.2.4.1.1.3 The command generation logic shall be optimized for TF control over rolling to moderately rough terrain. The system shall provide adequate control to protect the system from "clobber" during flights over all types of terrain, and shall emphasize minimum deviation from the commanded altitude over prominent peaks in the terrain.

3.4.2.4.1.1.4 The system shall provide approximately symmetrical clearance response in negotiating tower type obstructions on otherwise flat terrain, passing over the obstruction in nominally level flight within the limits specified below. This characteristic shall be maintained independent of obstruction height (for heights less than 3000 feet (914.4 m)) and constant vehicle speed of mach 0.4 to mach 0.8.

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Set Clearance	Undershoot	Overshoot
2000 feet (609.6 m)	600 feet (182.88 m)	0
1500 feet (457.2 m)	400 feet (121.92 m)	0
1000 feet (304.8 m)	200 feet (60.96 m)	0
500 feet (152.4 m)	50 feet (15.24 m)	50 feet (15.24 m)
200 feet (60.96 m)	50 feet (15.24 m)	50 feet (15.24 m)

3.4.2.4.1.1.5 Simulation over terrain samples (Cornell Aeronautical Laboratories Model 6201, and NOTS course 10 as a minimum) shall be performed at speeds of mach 0.4 and 0.8 and altitudes of 200 and 2000 feet (60.96 and 609.6 m) to demonstrate the terrain following performance. This simulation shall include all important error contributions and system limitations and constraints and shall assume vertical gusts of 4 ft/sec (1.219 m/sec) rms velocity.

3.4.2.4.1.2 Terrain Following System Constraints - The aircraft design constraints for the performance specified in paragraph 3.4.2.4.1.1 shall be as follows:

3.4.2.4.1.2.1 Dive Angle not to exceed 25 degrees (0.436 rad)

3.4.2.4.1.2.2 Positive g-limit: 2.0 incremental

3.4.2.4.1.2.3 Negative g-limit: 0.85 incremental

3.4.2.4.1.3 Vehicle Description - The description of the aircraft transfer function shall be supplied to the radar set manufacturer.

3.4.2.4.1.4 Turn Lead - In order to provide proper and adequate target and terrain illumination in a turn, the radar shall incorporate necessary lead information during turns. When turn rate exceeds 0.75 degrees (0.013 rad) per second, provisions shall be made to cause a fail signal as specified in paragraph 3.4.4.5.

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3.4.2.4.1.5 Variable Velocity - In order to obtain optimum terrain following performance, the radar shall make provisions for varying command computations as a function of aircraft groundspeed between mach 0.4 and 0.8.

3.4.2.4.1.6 Radar Altitude Control - The aircraft radar altitude signal described in paragraph 3.4.9.5 herein shall be utilized to provide an override function to the climb/dive command to improve flight safety in the event of undetected failure of the air navigation computer.

3.4.2.4.2 Antenna Beam - The antenna beam shall be an elevation monopulse as specified in paragraph 3.4.20.2.2.

3.4.2.4.3 Antenna Scan - The antenna beam shall be programmed in an elevation pattern which is turn rate, roll, pitch, and drift stabilized. The elevation scan coverage, limits and stabilization shall be as described in paragraph 3.4.20.4.

3.4.2.4.4 Range Coverage - A range of 10 sea miles shall be provided.

3.4.2.4.5 Radar Indicator Scope Display - The display in the terrain following mode shall be presented on the radar indicator as an E^2 type of presentation as specified in paragraph 3.4.3.5.

3.4.2.4.6 Command Steering and Warning Output - The radar set shall provide signals as defined in paragraphs 3.4.10.8 and 3.4.10.9 to drive the Attitude Director Indicator and Head-Up Display in terrain following modes.

3.4.2.4.7 Terrain Clearance - When the terrain following mode is selected, the terrain clearance setting shall be selectable at discrete steps of 200, 500, 1000, 1500, and 2000 feet (60.96, 152.4, 304.8, 457.2 and 609.6 m).

3.4.2.4.8 Weather Rejection - The weather return rejection feature as described in paragraph 3.4.5 shall be provided in the terrain following mode.

3.4.2.5 Terrain Following/Terrain Avoidance - A cross-scan terrain aviodance (CSTA) mode shall combine the requirements of paragraphs 3.4.2.3 and 3.4.2.4 except that the azimuth scan limits shall be a minimum of $\pm$ 20 degrees (0.349 rad). In this mode, the azimuth and elevation scans shall be time shared as specified in paragraph 3.4.20.4 to generate both terrain following and terrain avoidance information during the scan cycle. Terrain following information shall be generated during the elevation scan and the flight command signal provided as specified in paragraph 3.4.2.4, except that the flight command shall command the aircraft within $\pm$ 20 feet

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(6.096 m) of the altitude commanded in the TF mode over level terrain. Terrain Avoidance information shall be generated during the azimuth portion of the scan and displayed on the indicator in accordance with paragraphs 3.4.3.1 and 3.4.3.4.

3.4.2.6 Terrain Following/Ground Mapping (Pencil) - A terrain following, Ground Mapping (CSGMP) mode shall combine the requirements of paragraphs 3.4.2.2 and 3.4.2.4 except the ground mapping azimuth scan limits shall be a minimum of ± 20 degrees (0.349 rad) and the range scales shall be limited to 5 and 10 sea miles. Provision shall be made to automatically limit the range scale to ten sea miles if a greater range is selected. In this mode, the antenna elevation and azimuth scans are time-shared as specified in paragraph 3.4.20.4. Terrain following information shall be acquired during the elevation scan and flight commands shall be provided as specified in paragraph 3.4.2.4 except that the flight command shall command the aircraft within ± 20 feet (6.096 m) of the altitude commanded in the TF mode over level terrain. Ground map information shall be generated during the azimuth sector scan and displayed on the radar indicator. Display accuracy shall be as specified in paragraphs 3.4.3.1 and 3.4.3.2. Antenna elevation tilt control shall be provided for the azimuth portion of the scan as specified in paragraph 3.4.2.7.2, except that the antenna scan limits and servo response characteristics as specified herein shall not be exceeded.

3.4.2.7 Air-to-Ground Ranging Mode - The radar set shall provide the capability of ranging against a distributed (clutter) or an isolated (point) target with low background reflectivity in the AGR mode. This mode shall provide an output of range along the antenna boresight as defined in paragraph 3.4.10.9 and a discrete lock-on signal as defined in paragraph 3.4.10.5 to a Navigation Weapon Delivery Computer. A target reject as described in paragraph 3.4.9.24 shall cause the range tracker to unlock and search in range for another target. Upon selection of the AGR mode, the range tracker shall commence a search in range from 60,000 feet (18.294 km) to 1,400 feet (426.72 m) and lock-on the first target of sufficient strength. Failure to lock on prior to reaching 1400 feet (426.72 m) shall result in the range tracker recycling to 60,000 feet (18.294 km) and starting another search cycle. In the air-to-ground ranging mode, the antenna shall be referenced as defined in paragraph 3.4.20.9. The antenna shall be either manually adjustable about this reference in elevation as specified in paragraph 3.4.2.7.2 and driven to boresight in azimuth, or shall be slaved to azimuth and elevation signals from an external source as specified in paragraph 3.4.2.7.1.

3.4.2.7.1 Antenna Slaving - Upon receipt of a discrete +28 volt dc signal, designated ANTENNA SLAVE as defined in paragraph 3.4.9.13, the antenna shall be disconnected from the manual tilt control and azimuth boresight position, and slaved in azimuth and elevation to externally derived signals as defined in paragraph 3.4.9.14 with an accuracy as specified in paragraph 3.4.20.5.

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3.4.2.7.2 Manual Tilt Control - A manual tilt control shall be provided to manually point the radar beam in elevation. The angular position of the antenna boresight shall be capable of being positioned by means of the manual control to the desired setting through a range of +5 degrees (0.087 rad) to -15 degrees (-0.262 rad) about the boresight line with the accuracy specified in paragraph 3.4.20.5. A detent position shall be provided on the manual tilt control, which shall cause the antenna to be driven to the boresight line within the boresight alignment accuracy as specified in paragraph 3.4.20.6.

3.4.2.7.3 Antenna Beam - The antenna beam in air-to-ground ranging modes shall utilize elevation monopulse patterns as specified in paragraph 3.4.20.2.2.

3.4.2.7.4 Tracking Rate and Accuracy -

3.4.2.7.4.1 Tracking Rate and Accuracy (Clutter Targets) - The radar shall be capable of tracking terrain targets at range rates of at least 4000 feet (1219.2 m) per second. The range accuracy independent of the pointing error as specified in paragraph 3.4.20.5 at range rates of 4000 feet (1219.2 m) per second or less is specified below:

Reflectivity	Grazing Angle	Slant Range in Feet (meters) for ± 25-Foot (7.62 m) Accuracy (1σ)	Slant Range in Ft. (meters) for Accuracy of 0.25 Percent (1σ)
-20 dB	4 degrees (0.070 rad)	1400 to 10,000 (426.7 to 3048)	10,000 to 22,000 (3048 to 6706)
-20 dB	6 degrees (0.105 rad)	1400 to 10,000 (426.7 to 3048)	10,000 to 35,000 (3048 to 10668)
-20 dB	8 degrees (0.140 rad)	1400 to 10,000 (426.7 to 3048)	10,000 to 45,000 (3048 to 13716)
-20 dB	10 to 70 degrees (0.175 to 1.222 rad)	1400 to 10,000 (426.7 to 3048)	10,000 to 60,000 (3048 to 18288)

3.4.2.7.4.2 Tracking Rate and Accuracy (Isolated Targets) - For isolated targets, the radar shall be capable of tracking targets of a 100 square meter radar cross section or more at tracking rates up to 4000 feet (1219.2 m) per second independent of depression angle with the following accuracy:

3.4.2.7.4.2.1 ± 25 feet (7.62 m) from 1400 feet (426.72 m) to 10,000 feet (3048 m).

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3.4.2.7.4.2.2 ± 0.25 percent from 10,000 feet (3048 m) to 60,000 feet (18288 m).

3.4.2.7.5 Lock-on and Range Coverage - The 10-sea mile scale shall be used. The range tracker shall maintain track out to 60,000 feet (18288 m) at grazing angles above 10 degrees (0.175 rad) for clutter targets; otherwise, lock-on shall be maintained at ranges specified in paragraphs 3.4.2.7.4.1 and 3.4.2.7.4.2 under the conditions specified.

3.4.2.7.6 Radar Indicator Display - The display shall be as specified in paragraph 3.4.3.7.

3.4.2.7.7 Range Set and Readout - The air-to-ground ranging modes shall provide the capability for the pilot to set a range on the set control and obtain an indication from the radar when the range has been achieved. The accuracy and resolution with which the range can be set and the indication furnished shall be in accordance with paragraph 3.4.22.1.

3.4.2.8 Beacon Mode - The beacon mode shall cause the radar set to transmit appropriate pulse modulated RF signals to interrogate an AN/APN-134 rendezvous beacon installed in an airplane, and receive responses from the beacon at ranges up to 80 sea miles. The beacon responses shall be displayed on a depressed center PPI presentation. The beacon interrogation and response characteristics shall be as specified in paragraph 3.4.2.8.6.

3.4.2.8.1 Antenna Function - The antenna beam shall be a pencil beam formed by the elevation monopulse sum pattern in the beacon mode.

3.4.2.8.2 Antenna Scan - The antenna shall scan ± 45 degrees (0.785 rad) in azimuth, centered about the ground track. Antenna scan coverage, limits, and stabilization shall be as specified in paragraph 3.4.20.4.

3.4.2.8.3 Range Coverage - Range scales of 5, 10, 20, 40, and 80 sea miles shall be selectable by the pilot.

3.4.2.8.4 Radar Indicator Scope Display - The coded beacon video returns shall be displayed in slant range and azimuth coordinates on the radar indicator in a depressed center PPI presentation. The accuracy of the range and azimuth of the first beacon return shall be as specified in paragraph 3.4.3.1 and 3.4.3.2.

3.4.2.8.5 Manual Tilt Control - Manual tilt control as specified in paragraph 3.4.2.7.2 shall be provided in the beacon mode.

3.4.2.8.6 Beacon Characteristics (AN/APN-134) - The AN/APN-134 characteristics are as follows:

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3.4.2.8.6.1 Interrogation Pulse Requirements

a. RF Frequency: 16,500 ± 40 megahertz

b. Pulsewidth: 2.0 ± 0.5 microseconds (half amplitude)

c. Receiver Sensitivity: -66 dBm

d. Pulse repetition frequency (PRF) limits: Receiver blanked 200 ± 40 microseconds after interrogation.

3.4.2.8.6.2 Response Characteristics

a. RF Frequency: 16,280 ± 2 megahertz

b. Pulsewidth: 0.75 ± 0.25 microseconds (half amplitude)

c. Pulse Code: 6 maximum filled positions out of 9 possible, spaced at 6 ± 1 microseconds

d. Maximum Duty Cycle: 0.003

e. Peak Beacon Power: 2 kW

f. Antenna Pattern: Assume unity gain

g. Transponder Delay: 1.5 ± 0.5 microseconds

3.4.2.8.7 Antenna Polarization - Antenna polarization shall be circular.

3.4.2.9 Television Display Mode - The radar system shall have the capability to accept a television signal, with the characteristics specified in paragraph 3.4.9.15, and to display it on the Radar Indicator as specified in paragraph 3.4.3.3. Automatic switching shall occur upon receipt of a discrete ground signal indicating the presence of a TV signal. The radar set shall continue to operate in the selected mode except for the interruption of the display. When terrain following, or a mode combined with terrain following is selected, a climb/dive command shall be available in the output for driving the horizontal pointer of the Attitude Director Indicator. An override switch shall be provided to remove the TV picture from the display and present the normal display for the programmed mode. A light shall be provided to indicate an available TV signal. TV Display shall be available in radar standby mode after time-out.

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3.4.2.10 SIDS Display Mode - The radar system shall have the capability of displaying SIDS information on the radar indicator. Automatic switching shall occur on the receipt of a discrete signal. The SIDS display shall preempt any other display except the TV display. The override switch referred to in paragraph 3.4.2.9 shall also perform an override function for the SIDS display. A light shall be provided to indicate an available SIDS signal. SIDS Display shall be available in radar standby mode after time-out.

3.4.2.11 RHAW Display Mode - No Radar Homing and Warning (RHAW) display functions are required; however, space shall be allocated for future possible RHAW incorporation. This space shall consist of the following minimum provisions:

3.4.2.11.1 Space provisions in the Set Control for a RHAW mode pushbutton switch.

3.4.2.11.2 One twenty-gauge pin on a Set Control connector for mode logic to the Sweep Generator.

3.4.2.11.3 One twenty-gauge pin on a Sweep Generator connector for RHAW mode logic to the Set Control.

3.4.2.11.4 Four coaxial pins on a Sweep Generator connector.

3.4.2.11.5 Space to be provided on the RHAW/SIDS module board in the Sweep Generator for RHAW circuits incorporation.

3.4.2.12 Additional Display Capabilities - The radar set, when configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V), will have the capabilities described in paragraphs 3.4.2.12.1 through 3.4.2.12.3.

3.4.2.12.1 FLIR Display Mode - When the appropriate input commands are supplied to the radar set, the display will be capable of displaying 875 line FLIR video information.

3.4.2.12.2 Expanded Map Display - When used with a modified power supply PP-6130/APQ-126(V), the radar set will have an "expand about cursor intersection" (EXPAND) capability.

3.4.2.12.3 Hold Display - The radar set will provide the capability of inhibiting memory up-dating in the HOLD mode.

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3.4.3 Display Requirements

3.4.3.1 Display Limits and Accuracy - The range sweep shall follow the position of the antenna within ± 0.5 degrees (8.7 mrad) in mapping, beacon, and TA modes of operation. The range sweep start shall be positioned within 1/16 inch (15.875 mm) of the zero range and the maximum sweep range shall coincide with the maximum range scale as scribed on the overlay within ± 2 percent. The display shall be made on a 5-inch tube (12.7 cm) with a minimum viewable diameter of 4 inches (10.16 cm). The sweep deviations from a straight line shall be within ± 0.5 degree (8.7 mrad) accuracy requirement. It shall be possible, with proper indicator adjustments, to obtain a spot size no greater than 10 mils (0.254 mm).

3.4.3.2 PPI Display - The displayed ground range shall have a maximum error of ± 2 percent of the maximum displayed range exclusive of error introduced by the use of the altitude signal as specified in paragraph 3.4.9.6 for delay of the sweep start.

3.4.3.3 TV Display - Capability shall be incorporated in the radar system to accept a composite TV signal consisting of video, horizontal, and vertical sync pulses and displaying same on a minimum four inches (10.16 cm) of the 5-inch (12.7 cm) display tube. The resolution shall be at least 400 vertical lines for a 500 line input. Six shades of grey shall be available under ambient cockpit lighting. Lighting shall be provided on the Indicator to notify the pilot when TV or SIDS is available for display.

3.4.3.4 Terrain Avoidance Display - The display circuit mechanizations shall provide a display at the correct range and azimuth of all video which represents terrain protruding above the aircraft's altitude. Terrain which does not extend within 100 feet (30.48 m) of that plane shall not be displayed within a range of 2.5 sea miles. Between 2.5 and 10.0 sea miles, this tolerance shall increase proportionally to 400 feet (121.92 m) at 10.0 sea miles.

3.4.3.5 E^2 Scan Display - In the terrain following mode, the display circuits shall be mechanized to display the above boresight video on rectangular coordinates. The antenna elevation angle shall be a linear representation on the vertical scale. The range shall be approximately logarithmic representation on the horizontal scale with a maximum range of 10 sea miles. The accuracy of the display shall be consistent with range and elevation accuracy of the video information. The aircraft command reference line shall also be presented on the display.

3.4.3.6 Air-to-Ground Range Display - The air-to-ground range display shall be a single vertical trace centered horizontally on the scope. The range to the target shall be shown by horizontally deflecting the sweep at the range of the terrain along boresight. Maximum range shall be 10 sea miles.

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3.4.3.7 Display Tube Characteristics (IP-940/APQ-126(V) and IP-952/APQ-126(V)) - The display tube shall be a P-4 phosphor type direct view storage tube (DVST) and shall have a minimum resolution of 100 shrinking raster lines per inch with 1000 feet lamberts brightness when displaying TV. The erase time of the storage tube shall be no greater than 1 millisecond. A storage time of up to three seconds, adjustable by the pilot, shall be provided.

3.4.3.8 Display Tube Characteristics (IP-1256/APQ-126(V)) - The display tube must have at least 4 inches (10.16 cm) of useable display, resolution of 600 horizontal elements (minimum) for a 700 element input and three shades of gray contrast under cockpit ambient lighting conditions of 10,000 footlamberts. The tube shall have a resolution of at least 185 shrinking raster lines per inch with 1000 foot lamberts brightness at the face of the tube.

3.4.4 Failure Monitoring and Indication -

3.4.4.1 Radar Fail Indication - An indication in the form of a warning light on the radar indicator, loss of excitation to the horizontal pointer flag alarm, and a full climb command to the horizontal pointer of the ADI also causing pullup command (flashing x) and full climb of the flight director on the Head Up Display shall be provided to alert the pilot when the radar set operation is unsafe in all terrain following modes.

Detection of unsafe radar performance by any of the failure monitors shall result in the following:

RADAR DISPLAY	MODES
Fail Indicator Light "ON"	TF, CSTA, CSGMP, TA
Excitation Removed from Fail Flag	TF, CSTA, CSGMP
Full Climb Command to Horizontal Pointer	TF, CSTA, CSGMP

In order to detect unsafe operation, the functions to be continuously monitored shall include but are not limited to the following:

3.4.4.1.1 Transmitter Power - The transmitter power shall be continuously monitored in the terrain avoidance, terrain following, and cross-scan modes of operation.

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3.4.4.1.2 Receiver Automatic Frequency Control - The performance of the receiver automatic frequency control shall be continuously monitored in the terrain following, terrain avoidance, and cross-scan modes of operation.

3.4.4.1.3 Antenna Elevation Scan - The antenna elevation scan shall be continuously monitored in the terrain following and cross-scan modes of operation.

3.4.4.1.4 Receiver-Computer Response - The receiver and command computer response shall be monitored during each antenna scan in the elevation plane in the terrain following and cross-scan modes. The receiver-computer shall be monitored by injecting a pulse of radio frequency (RF) energy into the waveguide assembly as a calibrated target return and comparing the command computer response with a preset reference to obtain an end-to-end check of radar operation.

3.4.4.1.5 Test Pulse (BIT) Response - The test pulse circuitry shall be monitored during each antenna scan in the elevation plane in the terrain following and cross-scan modes of operation.

3.4.4.1.6 Servo Repeater (BIT) Response - The servo repeater circuitry shall be continuously monitored in the terrain-following and cross-scan modes of operation.

3.4.4.2 Radar Altimeter Fail Indication - An indication in the form of a full climb command to the horizontal pointer of the Attitude Director Indicator and loss of excitation to the horizontal needle flag alarm shall be provided in terrain following and cross-scan modes to alert the pilot when the Radar Altimeter fail signal as specified in paragraph 3.4.9.7 indicates an unreliable signal.

3.4.4.3 Heading and Attitude Reference System Fail Indication - An indication in the form of a full climb command to the horizontal pointer of the Attitude Director Indicator and loss of excitation to the horizontal pointer flag alarm shall be provided in Terrain Following and Cross-Scan modes to alert the pilot when the Heading and Attitude Reference fail signals as specified in paragraph 3.4.9.8 indicate an unreliable signal.

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3.4.4.4 Weapon Delivery Computer Fail Indication - An indication in the form of a full climb command to the horizontal pointer of the Attitude Director Indicator and loss of excitation to the horizontal needle flag alarm shall be provided in Terrain Following and Cross-Scan modes to alert the pilot when the velocity, drift, and flight-path angle inputs from the computer are unreliable.

3.4.4.5 Excessive Turn Rate - An indication in the form of a full climb command to the horizontal pointer of the Attitude Director Indicator and a loss of excitation to the horizontal pointer flag alarm shall be provided if turn rate signal exceeds 0.75 degrees (13.1 mrad) per second.

3.4.4.6 Inertial Measurement Set, Failure Indication - An indication in the form of a full climb command to both the Head Up Display and Attitude Direction Indicator, loss of horizontal pointer alarm flag excitation, full up command on the Head Up Display shall be provided in the TF and cross-scan modes to alert the pilot of an inertial platform failure.

3.4.5 Weather Return Rejection - The radar system shall be capable of decreasing energy return from rain at the option of the pilot as specified in paragraph 3.4.21 for all modes using the monopulse antenna patterns, except the Beacon and AGR modes by means of circular polarization. Circular polarization only will be used in the Beacon mode, while linear polarization only will be used in the AGR mode. The antenna shall provide a minimum integrated cancellation ratio of 15 dB when measured at zero-degree elevation scan.

3.4.6 Mode Switching - The Radar Set Control and Aircraft Master Function Switch (and the Indicator IP-1256/APQ-126(V) Expand/HOLD/ Normal switch when configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V)) combined shall provide the capability of selecting the following modes:

a. Off

b. Standby (STBY)

c. Terrain Following (TF)

d. Terrain Avoidance (TA)

e. Terrain Following/Terrain Avoidance (CSTA)

f. Terrain Following/Ground Mapping (CSGMP)

g. Ground mapping, pencil beam (GMP)

h. Ground mapping, shaped beam (GMS)

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i. Beacon (BCN)

j. Air-to-Ground Ranging (AGR)

NOTE: No position is required for the TV display mode or Shrike Integrated Display System (SIDS), which shall be automatic selections.

The following modes are available when the radar set is configured with Indicator IP-1256/APQ-126(V) and Convertor CV-3368/APQ-126(V) and Power Supply PP-6130/APQ-126(V) has been appropriately modified. FLIR operation is dependent upon the necessary inputs.

k. Forward Looking Infrared Display (FLIR)

l. Expanded Map Display (EXPAND)

m. Frozen Map Display (HOLD)

NOTE: No position is required for the FLIR display mode which shall be automatic selection.

3.4.6.1 OFF Position - The OFF positon shall remove primary power from the system. The switch design shall include a mechanical interlock to prevent inadvertent turnoff.

3.4.6.2 STBY Position - The STBY position shall apply standby power to the system. The radar transmitter shall not be activated in STBY. Upon switching from STBY to any operational mode, the radar shall be immediately available, provided that the initial warmup time specified in paragraph 3.4.11 has been met.

3.4.7 Range Scale Selection - Provisions shall be made in the radar to change range scales upon a momentary switch closure external to the radar system. Sequencing shall be from maximum to minimum range. Each closure of the external switch shall provide a range change, with ranges for each mode as follows:

Radar Display	Range Selection
TF	10
TA	10, 5
CSTA	10, 5
CSGMP	10, 5

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Radar Display	Range Selection
GMP	80, 40, 20, 10, 5
GMS	80, 40, 20, 10, 5
AGR	10
EXPAND (when available)	2.5

3.4.8 Terrain Clearance Setting Control - The terrain clearance setting control, located on the Radar Set Control, shall be variable between 200 feet (60.96 m) and 2000 feet (609.6 m), with detented discrete clearance settings at 200, 500, 1000, 1500, and 2000 feet (60.96, 152.4, 304.8, 457.2 and 609.6 m).

3.4.9 Radar Set, Input Signals - The following signals shall be provided from aircraft systems under the interface conditions set forth in Table III.

3.4.9.1 Roll Signal - Signals shall be provided from a synchro representing the aircraft vertical roll reference through aircraft roll attitudes of 360 degrees (6.283 rad). The synchro shall be excited by phase C-0 degree aircraft power. The static accuracy of the input signal shall be ± 0.25 degree (4.4 mrad). The roll signal to the radar shall have the following characteristics:

Voltages	11.8 V, 400 Hz
Total null voltage	30 mV, 400 Hz
Fundamental null voltage	20 mV, 400 Hz
Transformation Ratio	0.45 ± 4 percent
Voltage Gradient	0.206 V/degree (11.803 v/rad)
Phase shift	8.0 degree lead (0.140 rad)

3.4.9.1.1 Rotation - For wings level (zero roll), the roll synchro stator-rotor voltage relationship shall be X to Z = Y to Z (null X to Y) and in phase with H to C. For counter-clockwise rotation of the antenna from the zero position (looking forward) X to Z increases before decreasing in phase with H to C.

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3.4.9.2 Pitch Signal - A signal from a pitch synchro shall be provided for aircraft pitch of at least ± 60 degrees (1.047 rad). The pitch synchro shall be excited by phase C aircraft power. The static accuracy of the input signal shall be ± 0.25 degree (4.4 mrad). The pitch signal to the radar shall have the following characteristics:

Voltages	11.8 V, 400 Hz
Total null voltage	30 mV, 400 Hz
Fundamental null voltage	30 mV, 400 Hz
Transformation ratio	0.45 ± 4 percent
Voltage gradient	0.206 V/degree (11.803 v/rad)
Phase shift	8.0 degrees lead (0.140 rad)

3.4.9.2.1 Rotation - For zero pitch, attitude (relative to aircraft boresight line) the pitch synchro stator-rotor voltage relationship shall be X to Z = Y to Z (null X to Y) and in phase with H to C. For pitch down of the antenna from zero position X to Z increases before decreasing in phase with H to C.

3.4.9.3 Flight-Path Angle Signal - A signal proportional to aircraft Flight-Path Angle from -30 to +30 degrees (-0.524 to +0.524 rad) shall be provided from the aircraft central computer system. Positive flight-path angle is defined as that angle the velocity vector makes above the horizontal plane. This signal will be transmitted in dc voltage analog form. The scale factor shall be 0.6 volt per degree (34.4 volts/rad) with an accuracy of ± 0.25 degree (4.4 mrad) (1σ).

3.4.9.4 Drift Angle Signal - A drift signal shall be provided from a synchro which is aligned to aircraft drift angles up to 10 degrees (0.175 rad). The synchro shall be excited by 26.0 volts ac at phase C-0 degrees. The static accuracy of the input signal shall be ± 0.25 degree (4.4 mrad) (3σ). The drift angle signal to the radar shall have the following characteristics:

Voltage	11.8 V, 400 Hz
Total null voltage	30 mV, 400 Hz
Fundamental null	20 mV, 400 Hz

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Transformation ratio	0.45 ± 4 percent
Voltage gradient	0.206 V/degree (11.803 v/rad)

3.4.9.4.1 Rotation - For zero azimuth position (relative to aircraft boresight line) the azimuth synchro stator-rotor voltage relationship shall be X to Z = Y to Z (null X to Y) and out of phase with H to C. For azimuth right point from zero position, X to Z increases before decreasing out of phase with H to C.

3.4.9.5 Radar Altitude Signal - A signal proportional to aircraft altitude above the terrain will be provided for altitudes from 0 to 5,000 feet (1524 m). The signal shall be supplied from a high impedance source to an accuracy of ± 7 feet (2.13 m) or 7 percent (whichever is greater). The signal shall be a linear voltage analog of altitude with a scale factor of five millivolts per foot (16.4 mv/m) from zero to 5,000 feet (1524 m) of altitude. The maximum altitude rate of the signal shall be no less than 400 feet (122 m) per second at altitudes below 400 feet (121.92 m) and no less than 3000 feet (914.4 m) per second above 400 feet (121.92 meters). The input impedance shall be a minimum of 100,000 ohms.

3.4.9.6 Sweep Delay Altitude Signal - A signal proportional to aircraft altitude (barometric) shall be provided for altitudes from -1000 to +50,000 feet (-304.8 m to 15240 m). The signal shall be supplied by a 20,000-ohm potentiometer with a scale factor of 0.5 millivolt per foot (1.64 mv/m). The linearity of the potentiometer shall be ± 1 percent maximum. The potentiometer shall be excited by a signal from the radar set. This signal is to be used for altitude reference except when a valid radar altitude signal, as indicated in paragraph 3.4.9.7, is present. A radar altitude signal shall be provided with a scale factor of 0.5 millivolt per foot (1.64 mv/m). The input impedance shall be a minimum of 100,000 ohms.

3.4.9.7 Radar Altitmeter Fail Signal - A signal shall be provided to indicate reliable operation of the radar altimeter. The signal shall be +5 volts dc for reliable operation and 0 volt for unreliable operation.

3.4.9.8 Inertial Measurement Set Fail Signal - A signal shall be provided to indicate the status of the inertial measurement set operation. The characteristics of the signal are: +28 volts dc for safe operation, and zero volt for unreliable operation.

3.4.9.9 Yaw Rate Input - A signal proporational to turn rate shall be provided from a 1000-ohm source, for turn rates up to ± 30 degrees (0.524 rad) per second. The scale factor shall be 167 mV/degrees/second (9568 v/rad/sec) ± 1 percent (1σ). The signal shall be in phase with Phase C aircraft power for a right turn maneuver. The phase shift shall be ± 8 degrees (0.140 rad) maximum over the temperature range from -55°C to 75°C. The input impedance shall not be less than 100,000 ohms.

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3.4.9.10 Ground Velocity Input - A 0 - to 18 - volt dc analog signal, into 100,000 ohms, proportional to aircraft velocity from 0 Kts to 900 Kts ± 1 percent shall be provided.

3.4.9.11 Terrain Following Mode Discrete Signal - A discrete signal consisting of +28 volts dc shall be provided to the radar from the Master Mode Select switch when the Master Mode Select switch is in the TF position. This signal shall be used in conjunction with the Radar Control Selector switch functions to select the appropriate TF, TA, and Ground Map cross-scan modes.

3.4.9.12 Antenna Slaving Discrete - A discrete +28-volts dc signal shall be provided to initiate antenna slaving in AGR modes and maintain it as long as required when certain armament delivery modes are initiated. Upon receipt of this command, the radar shall go into the AGR mode.

3.4.9.13 Antenna Slaving Signals -

3.4.9.13.1 Elevation Slaving Signal - A 3-wire synchro signal shall be provided from an external source to which the antenna shall be slaved in air-to-ground ranging modes during the presence of the discrete signal specified in paragraph 3.4.9.12. The synchro shall be excited by the aircraft 0 degree Phase C power. The range of signal shall be ± 20 degrees (0.349 rad) of elevation angle from the aircraft elevation reference (boresight line as defined in paragraph 3.4.20.10).

The elevation slaving signal shall have the following characteristics:

Voltage	11.8 V, 400 Hz
Total null voltage	30 mV, 400 Hz
Fundamental null	20 mV, 400 Hz
Transformation ratio	0.45 ± 4 percent
Voltage gradient	0.206 V/degree (11.803 v/rad)

Rotation - X to Z increases from decreasing in phase with H to C for antenna down command from zero position referenced to aircraft boresight line.

3.4.9.13.2 Azimuth Slaving Signal - The azimuth slaving signal shall have a range of ± 40 degrees (0.698 rad). It shall have the following characteristics:

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Voltage	11.8 V, 400 Hz
Total null voltage	30 mV, 400 Hz
Fundamental null	20 mV, 400 Hz
Transformation ratio	0.45 ± 4 percent
Voltage gradient	0.206 V/degree (11.803 v/rad)

Rotation - X to Z increases before decreasing out of phase with H to C for antenna right command from zero position referenced to aircraft boresight line.

3.4.9.14 TV Available - A ground signal shall be provided to the radar when a weapon station bearing a TV weapon is selected.

3.4.9.15 TV Video - A composite video signal shall be provided in conjunction with paragraph 3.4.9.14, the characteristics of which shall comform to WS-3180A, dated 26 November 1965.

3.4.9.16 AC Power Inputs - The following inputs of ac power shall be provided for reference use in the radar set.

Voltage	Phase	VA
26 volts ac	θC-90 degree (1.571 rad)	50
26 volts ac	θC	15
11.8 volts ac	θC	5
28.6 volts ac	θC	200

3.4.9.17 Computer Fail Signal - A signal shall be provided to indicate the status of the Weapon Delivery Computer operation. The characteristics of the signal are: 28 Vdc for safe operation, and 0 Vdc for unreliable operation. Detection of unreliable operation shall result in the interruption of the drift input signal. The antenna scan shall be referenced to the azimuth boresight position and the digital word outputs shall be referenced to an internal 50 kHz clock.

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3.4.9.18 Computer Driven Cursor Enable - A discrete 28 volts dc signal shall be provided to the radar to disable the manual range cursor defined in paragraph 3.4.2.1.1, and to enable the range and azimuth cursors defined in paragraph 3.4.2.1.6. Automatic selection of ground map (pencil) mode shall be coincident with this discrete unless ground map (shaped) is selected on the radar set control in which case GMS shall have precedence.

3.4.9.19 Cursor Position Data - Range and azimuth to a fixed point on the ground will be provided to position the range and azimuth cursors defined in paragraph 3.4.2.1.6.

3.4.9.20 Missile Direction Finding - Internal switching shall be provided to switch the display to the SIDS mode upon receipt of the discrete signal described in paragraph 3.4.9.22. When in the SIDS mode, the radar indicator shall accept and display target information provided as auxiliary inputs from an external source. Inputs to be supplied are horizontal and vertical deflection and a write discrete signal.

3.4.9.21 Missile Direction Finding Discretes -

3.4.9.21.1 SHRIKE Selected - When a weapon station carrying a SHRIKE missile has been selected, a ground will be provided to the radar system. When a SHRIKE station is not selected, the line will be an open circuit.

3.4.9.22 Range Select - To accomplish the radar range scale selection as defined in paragraph 3.4.7, an external switch closure shall be provided. Maximum current through the switch shall not exceed 1 ampere.

3.4.9.23 Target Reject - An external switch closure shall be provided to enable the radar to break lock-on when in the AGR mode. Maximum current through the switch shall be 1 ampere.

3.4.9.24 AGR Ready Discrete - Upon receipt of this discrete, the radar shall be placed in a non-radiating AGR Ready mode. This condition is to be overridden when the Antenna Slaving discrete described in paragraph 3.4.9.12 is present.

3.4.9.25 FLIR Video - A composite video signal shall be provided in conjunction with the signal described in paragraph 3.4.9.26, the characteristics of which shall conform to EIA-STD RS-343-A.

3.4.9.26 FLIR Available - A ground signal shall be provided in conjunction with the signal described in paragraph 3.4.9.25 when a FLIR display is desired.

3.4.10 Output Signals - The radar set shall provide the following output signals under the interface conditions set forth in Table II.

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3.4.10.1 Terrain Clearance Command Discrete - A discrete signal output will be provided from the radar for the 200-foot (60.96 m) terrain clearance command altitude selection. A +28 volt dc signal shall indicate that the 200-foot (60.96 m) terrain clearance command has been selected; a 0 volt dc signal shall indicate that command is not selected.

3.4.10.2 Antenna Scan Reverse Signal -

3.4.10.2.1 Synchronization - The antenna scan reversal pulse synchronization shall be as follows:

a. The leading edge of the pulse shall occur at maximum elevation scan in terrain following modes.

b. The leading edge of the pulse shall occur on completion of azimuth portion of cross-scan modes.

c. The leading edge of the pulse shall occur in completion of each right-to-left scan in terrain avoidance, ground map, and beacon modes.

3.4.10.2.2 Synchronization Accuracy - The synchronization accuracy of the pulse shall be coincident $\pm$ 15 milliseconds with the end of scan specified in paragraph 3.4.10.2.1.

3.4.10.2.3 Pulsewidth - The pulsewidth of the antenna reversal pulse shall be 15 $\pm$ 3 milliseconds measured at the 50 percent amplitude point.

3.4.10.2.4 Rise Time - The rise time of the antenna reversal pulse shall be 1.0 millisecond maximum from the 3.0 to 10.0 volt amplitude level.

3.4.10.2.5 External Loading - External loading by airplane avionics will be no less than 15 kilohms resistive.

3.4.10.2.6 Amplitude - The voltage amplitude of the antenna reversal pulse shall be 15 volts minimum measured at the midpoint of the pulsewidth.

3.4.10.3 Radar Altimeter Fail Signal - A ground signal shall be provided from the radar system when the +5 volt dc safe signal described by 3.4.9.7 is not present for use in switching external equipment.

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3.4.10.4 Altitude Potentiometer Excitation - Excitation voltage shall be provided to the barometric altitude potentiometer to supply the input specified in paragraph 3.4.9.6.

3.4.10.5 Radar Range Lock-On - The radar set shall provide a signal to the Navigation Weapon Delivery Computer to signify that the radar is locked-on in range in the AGR mode. The signal shall be transmitted as a validity bit in the range word.

3.4.10.6 Blanking Pulse - The radar shall provide a blanking pulse of +15 ± 3 volts in a 93 ohm load. The leading edge of the blanking pulse shall precede the transmitter pulse by 0.5 microsecond minimum and shall be sufficiently long to blank the transmitted pulse for all pulse repetition frequencies.

3.4.10.7 Climb/Dive Command Signal - The radar set shall provide a flight command signal suitable for driving the horizontal pointer of the attitude director indicator and elevation maneuvering command (Zc) as required by the head-up display in all terrain following modes. The signal characteristics shall be as required by Specification MIL-I-23524 and shall be fully compatible with the terrain following requirements of paragraph 3.4.2.4 of this specification. The HUD input impedance of this function shall not be less than 100,000 ohms.

3.4.10.8 Flight Command Warning Signal - The radar set shall provide a warning signal suitable for driving the horizontal pointer flag alarm of the attitude director indicator in all TF modes. In the event of a radar failure, a computer failure, a radar altimeter failure, or an inertial measurement set (IMS) failure this signal shall be interrupted to allow the fail flag to come into view. The flag shall be out of view when a polarized direct current of 245 microamperes is applied. The signal shall not exceed 380 microamperes. The impedance at the input terminals shall be 1000 ohms ± 3 percent.

3.4.10.9 Radar Range - The radar set shall provide an output of radar range. The range covered shall be 1400 to 60,000 feet (426.72 to 18288 m). The accuracy of the range output shall be as specified in paragraph 3.4.2.7.4. These outputs shall be in a digital format compatible with the Navigation Weapon Delivery Computer implementation. The digital range word transmission shall be in accordance with paragraph 3.4.10.9.1.

3.4.10.9.1 Digital Data Transmission - Digital interface electronics shall provide for a continuous serial transfer of data to and from the computer. The form of the signals shall meet the following requirements:

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3.4.10.9.1.1 Data Word - Data word shall be transmitted over a data channel. A data channel shall consist of the following:

a. One pair of data lines consisting of true and ones complements.

b. One pair of lines for clock pulses.

c. One pair of data ready lines as defined in paragraph 3.4.10.9.1.1.1.

d. One Signal return line.

e. One pair of data word identity lines.

3.4.10.9.1.1.1 Data Ready Signal - The data ready signal shall be one-bit length in duration and shall occur concurrent with bit number 1 of the data word. The leading edge of the ready signal shall be concurrent with the leading edge of the clock pulse.

3.4.10.9.1.2 Data Word Format - The digital data word format shall consist of a 20-bit word. The bit designations shall be as follows:

Bit	Designation
1	Control bit
2	Validity bit
3 - 18	Data bits including sign (in order of increasing weight)
17	MSB of data
18	Sign bit of data
19	Spare bit (digital zero)
20	Parity bit (odd)

The transmission sequence shall be such that the control bit is transmitted first. When parmeter range on resolution does not require all 16 data bits, bit number 17 shall be the most significant bit, followed by descending bit order; bit number 18 shall be the sign bit; and zeros shall be inserted into the remaining least significant data bits beyond the resolution requirements specified in this specification.

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3.4.10.9.1.3 Data Word Identification - The data word identification shall be transmitted simultaneously with each data word and serves to identify each word. The format of this transmission shall be as follows:

Bit	
1	Control
2 - 13	Unused (digital zero)
14	Data identification (LSB)
15 - 18	Data identification (other identity bits)
19	Spare (digital zero)
20	Parity bit (odd)

3.4.10.9.1.4 Synchronization - The clock pulses will be provided by the central computer at a rate of 50kHz ± 10 percent and duty cycle of 50 percent. The pulses will be provided continuously. In the event the aircraft central computer shall fail, the radar shall be capable of providing air-to-ground range information to the Head-up display through the use of an internal backup clock.

3.4.10.9.1.5 Signal Characteristics - Pulse and data signals transmitted over a data channel shall have the following characteristics:

Rise time	Maximum 2 microseconds
Fall time	Maximum 2 microseconds
Digital (Binary) One State	0 ± 0.5 volts
Digital (Binary) Zero State	+5 ± 0.5 volts
Pulse coincidence	± 1.0 microsecond (entire data channel)
Sign Bit Pulse (+)	Logical Zero

3.4.10.9.1.6 Digital Data Receiver - A balanced differential circuit shall be used to receive the input true and complement digital signals from the navigational/weapon digital delivery computer. The receiver shall check for the presence of the control bit, and for odd parity and shall process the incoming data only if all of these checks are successful.

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3.4.10.10 Three-Phase Transformer Excitation - The radar shall provide 3-phase 115 volt ac power (reference phase A, B, C) to the external transformer from which the ac outputs of paragraph 3.4.9.16 shall be developed.

3.4.10.11 Terrain Following Mode Discrete - A discrete 28 volt dc signal shall be provided as an output when any mode which includes terrain following is selected.

3.4.10.12 Set Range Readout - The radar shall provide a 28 volt dc discrete signal when the range measurement reaches the value set on the range set control. This signal shall operate on an external light when the radar range is equal to, or less than, the preset value. The load capacity shall be for a 500 milliamperes resistive load.

3.4.10.13 Fail Signal - In the event of any failures defined in paragraph 3.4.10.8 herein, a +28 volt discrete signal shall be provided from the radar in all TF modes.

3.4.11 Time Delay - The radar set shall be fully operable within 200 seconds after the application of primary power to the system.

3.4.12 Output Power - The pulse power output of the radar transmitter shall not be less than 60 kilowatts peak as measured at the magnetron output flange.

3.4.13 Frequency - The frequency of the radar set shall be within a band of 16,500 $\pm$ 300 megahertz. Frequency of operation within this band shall be as specified in paragraphs 3.4.13.1 and 3.4.13.2 except that in the beacon and AGR modes, the transmitted frequency shall be 16,500 $\pm$ 40 megahertz.

3.4.13.1 Frequency Selectivity - Frequency of operation within the band specified shall be selectable at the option of the pilot, by means of a seven frequency selector switch in all modes except air-to-ground ranging and beacon modes.

3.4.13.2 Frequency Dither - A frequency dither about the selected frequency shall be provided in all modes except the Beacon mode. The dither shall be a minimum of 30 megahertz. The dither rate shall be 100 Hz $\pm$ 20 percent.

3.4.14 Receiver Performance - The receiver shall have a noise figure of not more than 10.0 dB and a receiver gain of 75.0 dB minimum.

3.4.15 Pulse Repetition Rate - The pulse repetition rates shall be 3600 + 0, -360 PPS for two-tenth-microsecond pulses, 720 +0, -72 PPS for 1-microsecond pulse, and 360 +0, -36 PPS for the 2-microsecond pulse.

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3.4.16 Pulse Duration - The duration of the transmitted pulse shall be 1.0 + 0.2, -0.2 microseconds as measured between the one-half power point for the 20-, 40-, and 80-mile ranges and shall be 0.2 + 0.04, -0.04 microseconds for all other ranges, except that in the beacon mode the pulse duration shall be within the limits specified in paragraph 3.4.2.8.6.

3.4.17 Receiver Bandwidth - The receiver bandwidth shall be as follows:

8 ± 1.6 MHz for 0.2 microsecond pulse modes

8 ± 1.6 MHz for beacon mode

1.6 ± 0.32 MHz for 1.0 microsecond pulse modes

3.4.18 System Sensitivity - The radar set tangential sensitivity shall be at least -86 dBm.

3.4.19 Sensitivity Time Control - Sensitivity Time Control (STC) shall be used to minimize signal amplitude variations from similar targets as a function of range.

3.4.20 Antenna Characteristics -

3.4.20.1 Antenna Size - The antenna reflector shall have overall dimensions not exceeding 13.3 inches (33.78 cm) in height and 20.1 inches (51.05 cm) in width.

3.4.20.2 Antenna Gain - A horizontally polarized pattern shall be provided for the GMS mode which has a minimum peak power gain of 28.5 dB. A minimum peak power gain of 28.5 dB shall be provided for the monopulse and GMP modes of operation.

3.4.20.2.1 Shaped Beam Antenna Function -

Polarization: Horizontal.

Elevation Pattern: $\csc^2 \theta \cos \theta + 2$ dB (-11-degree (-0.192 rad) to -60 degree (-1.047 rad) depression angle normalized at θ = -11 degrees (-0.192 rad)).

Peak power gain: 28.5 dB minimum.

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Azimuth Pattern: +3-dB beamwidth: -11-degree (-0.192 rad) depression angle, 2.5 degrees (.0436 rad) ± 0.25 degree (4.36 mrad); -40-degree (-0.698 rad) depression angle, 3 degrees (52.4 mrad) maximum; -60-degree (-1.047 rad) depression angle, 5 degrees (87.3 mrad) maximum.

Sidelobes: -11-degree (-0.192 rad) depression angle, 20 dB maximum
-60-degree (-1.047 rad) depression angle, 10 dB maximum.
(Relative to peak of beam at specified depression angle).

3.4.20.2.2 Antenna Monopulse and GMP Function -

Polarization - linear at 45 degrees (0.785 rad) or circular (selectable)

Sum Channel Elevation Function - consistent with performance requirements in all modes.

Peak power gain - 28.5 dB minimum

3.4.20.3 Monopulse On-Boresight Detection - Elevation monopulse on-boresight detection shall be used in the terrain following, terrain avoidance, and ranging modes. Either a linear or circularly polarized beam, in accordance with paragraphs 3.4 and 3.4.20.2 shall be used in the monopulse beams.

3.4.20.4 Antenna Scan - The radar set shall provide patterns of elevation scan, azimuth scan, and cross-scan (a time-shared combination of elevation and azimuth scans). The antenna scan patterns shall be stabilized per paragraph 3.4.20.8. The scan limits for each mode shall be as follows:

	Beam Center Scan Coverage	Beam Center Limits or Travel	Frame Time (Maximum)	Mode and Range
Elevation	+7°, -18° (+0.122,-0.314 rad)	+22°, -33° (+0.384,-0.576 rad)	0.25 second	TF, CSTA, CSGMP, 10 smi
Azimuth	+45° (48° max) (±0.785(0.838 rad max))	+55° (±0.960 rad)	0.55 second	TA, GMP GMS, 5, 10 smi
Azimuth	+20° (±0.349 rad)	+30° (±0.524 rad)	0.35 second	CSTA, CSGMPS, 5, 10 smi

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	Beam Center Scan Coverage	Beam Center Limits or Travel	Frame Time (Maximum)	Mode and Range
Azimuth	±45[a] (48° max) (±0.785(0.838 max)rad)	±55° (±0.960 rad)	1.10 second	GMP, GMS 20, 40 and 80 smi and Beacon 5, 10 20, 40, and 80 smi
Azimuth	±20° (±0.349 rad)	±30° (±0.524 rad)	1.10 second	EXPAND (when available)

3.4.20.5 Antenna Elevation Pointing Control - The radar shall be capable of pointing the antenna in response to slaving signals from an external source as specified in paragraph 3.4.2.7.1 during air-to-ground ranging modes only. The accuracy of the pointing, including thermal noise, shall be such that the elevation position of the antenna boresight shall be aligned with the angular position represented by the external signal within ± 0.3 degree (5.23 mrad) (1σ value) for tilt angles between +5 degrees (0.087 rad) and -15 degrees (-0.262 rad) and for zero roll and azimuth positions. The azimuth position shall be aligned with the pointing signal within ± 0.5 degree (8.73 mrad). For manual tilt, the elevation control range shall be +5 to -15 degrees (+87.2 mrad to 262 mrad) about the elevation reference line in the air-to-ground ranging, beacon, and ground mapping modes. The accuracy shall be ± 1 degree (17.5 mrad) from +5 to -15 degrees (+87.2 to 262 mrad) for manual tilt except in the ground map shaped mode in which the tilt accuracy shall be ± 3 degrees (52.4 mrad).

3.4.20.6 Antenna Mounting Details - The forward assembly radar mount shall include a mounting plate for the antenna which shall be at a nominal depression angle with respect to the mount of 1.5 degrees (26.2 mrad), and can be adjusted about the nominal position at least ± 1.5 degrees (26.2 mrad) in elevation to provide antenna alignment with the aircraft boresight line. The nominal position of the aircraft radar mounts is also at a depression angle of 1.5 degrees (26.2 mrad) with respect to the aircraft longitudinal reference.

3.4.20.6.1 Mechanical Boresight Requirements - Provisions shall be made for mounting a boresight fixture and optical sight on the antenna and adjusting the mechanical boresight of the antenna upon installation of the radar in the aircraft to the boresight line of the aircraft as defined in paragraph 3.4.20.10.

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3.4.20.6.2 Electrical Boresight Requirements - The antenna beam centerline shall be positioned within 0.5 degree (8.7 mrad) in azimuth and ±1 milliradian in elevation (1σ values) of the mechanical boresight telescope mounting pad. The beam centerline shall also be aligned in elevation with respect to the antenna mounting plate within ± 1 milliradian (1σ value).

3.4.20.6.3 Antenna Locking Provisions - Mechanical means shall be provided for locking the antenna in azimuth, elevation, and roll when the radar is turned off. The locking shall be accomplished by electro-magnetic brakes which are braked when de-energized.

3.4.20.7 Mechanical Boresight Repeatability - The angular displacement of the antenna boresight from its original position shall not exceed ± 2 milliradians in either azimuth or elevation when the forward assembly with its external cables and air connections is removed from its normal mounting in the airplane and then reinstalled in the same airplane.

3.4.20.8 Stabilization Limits - The antenna patterns shall be stabilized throughout the angular limits stated below. The radar shall not degrade the inherent accuracy of the various input quantities more than 0.1 degree (1.7 mrad) in any case.

	Beam Center Stabilization Limits (Minimum)	Static Accuracy of External Inputs
Pitch	± 15° (+7°, -18° scan stabilized to Pitch Angle Limits of +22°, -33°) (±0.262 rad (+0.122, -0.314 rad scan stabilized to Pitch Angle Limits of +0.384, -0.576 rad))	± 0.25° (4.4 mrad)
Roll Gimbal	± 45° (±0.785 rad)	± 0.25° (4.4 mrad)
Drift	± 10° (± 45° scan stabilized to limits of ± 55°) (±0.175 rad (±0.785 rad scan stabilized to limits of ±0.959 rad))	± 0.25° (4.4 mrad)

3.4.20.9 Antenna Reference -

3.4.20.9.1 Antenna Elevation Reference - The antenna zero pitch reference is along the aircraft boresight line and the elevation pointing signal is referenced to the aircraft boresight line.

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3.4.20.9.2 Antenna Azimuth Reference - The antenna azimuth position and the azimuth pointing signal shall be referenced to the aircraft boresight line.

3.4.20.10 Aircraft Boresight Line - The aircraft boresight line is defined as a line positioned at a -3 degree (-52.4 mrad) depression angle with respect to the longitudinal reference line, aligned to a zero (0) azimuth line.

3.4.20.11 Transmitter Dummy Load - A dummy load to terminate and absorb RF energy shall be incorporated as a part of the Antenna-Receiver unit.

3.4.21 Weather Rejection Feature Selection - Circular polarization shall be incorporated in the radar system as a means of reducing the energy from precipitation in all modes using the monopulse beams except AGR. This feature shall be made selectable by the pilot by a control on the control set.

3.4.22 Range Set Control - Controls shall be provided on the set control to allow position adjustment of the range cursor as specified in paragraph 3.4.2.1.1. The range position shall be continuously adjustable in increments of 0.1 sea mile from a range of zero to 19.9 sea miles on any selected map scale. A 3-digit readout shall be provided. In the AGR modes, this control shall also be used to set in a discrete range and to provide excitation to an external lamp, as specified in paragraph 3.4.10.12, when the radar range measured is equal to or less than the preset value. The range and accuracy shall be as specified in paragraph 3.4.22.1.

3.4.22.1 Accuracy and Resolution of Range Set and Readout - The accuracy of the range setting shall be within ± 100 feet (30.48 m) or ± 1 percent of the range output. The control shall be continuously variable in increments of 100 feet (30.48 m) from zero to 19,900 feet (6065.5 m).

3.4.22.2 Accuracy and Resolution of Manual Range Cursor - The accuracy of the manual range cursor defined in paragraph 3.4.2.1.1 shall be within ± 100 feet (30.48 m) or ± 1 percent of the range selected. The manual range control shall be variable in increments of 0.1 sea mile from zero (0) sea miles to 19.9 sea miles.

3.4.22.3 Accuracy of Driven Cursors - The driven cursors defined in paragraph 3.4.2.1.6 shall have the following accuracies:

3.4.22.3.1 The range accuracy shall be ± 100 feet (30.48 m) or ± 1/2 percent, whichever is greater, with respect to the commanded input.

3.4.22.3.2 The azimuth accuracy shall be ± 0.5 degree (8.7 mrad) with respect to the commanded input.

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3.4.23 Forward Assembly, Radar Group - The assembly shall consist of a group of radar set units integrated into a single package. Units of the group shall be as follows:

Antenna-Receiver
Power Supply-Programmer-Amplifier
Computer, Command
Transmitter, Radar
Mount, Forward Assembly

All units shall be interchangeable with a minimum of system readjustments. The scan programming function shall be accomplished by an all-electronic device rather than by an electro-mechanical device.

3.4.23.1 Mounting Provisions - The assembly shall be adequately protected to withstand the required vibration and shock environment when mounted to the airframe. The equipment shall be removable from the airframe as a complete assembly or by individual units. The design shall provide maximum access for system test points, adjustments and unit replacement from the forward end and sides of the package.

3.4.23.2 Electrical Interconnection - The assembly shall include a cable assembly which interconnects all wiring to the units thereon and provides for connecting the equipment package to an aircraft interconnecting harness.

3.4.23.3 Antenna Mounting Provisions - The assembly shall include a mounting structure for attachment of the units specified in 3.4.23 above. The mount shall include a mounting plate for the antenna which can be adjusted at least 1.5 degrees (26.2 mrad) to compensate for build-up tolerances and to provide antenna boresight alignment with the aircraft. Antenna boresight position is defined as the antenna beam centerline position in accordance with paragraph 3.4.20.10. In the standby mode the antenna shall be slaved to boresight. Mechanical means shall be provided for locking the antenna in the boresight position in azimuth, elevation, and roll when the radar mode switch is turned to "OFF". The locking shall be accomplished by electromagnetic brakes (stow clutches) which are braked when de-energized. Provisions shall be made for mounting a boresight fixture and scope on the antenna and adjusting the mechanical boresight as required for each installation.

3.4.23.4 Modulator Protection - A current sensing overload device shall be incorporated to remove input power from the high voltage power supply if excessive current is drawn by the modulator.

3.4.23.5 Crystal Protection - Receiver crystal protection in the form of a semi-conductor diode shall be provided to avoid crystal burnout in radar "OFF" and "STANDBY" modes. Twenty db of protection shall be provided.

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3.4.23.6 Magnetron Orientation - Any change involving the magnetron orientation or polarization shall be coordinated to eliminate possible aircraft environmental problems.

3.5 Detail Requirements -

3.5.1 Power Supply-Programmer, Antenna Scan, PP-6130/APQ-126(V) - The power supply-programmer shall meet the following requirements.

3.5.1.1 Function - The power supply-programmer shall contain:

3.5.1.1.1 Power Converting - The required circuits to convert the 115 volt, 400 Hz, 3-phase aircraft voltage to the ac and dc voltages required for system operation.

3.5.1.1.2 Servoamplifiers - Servoamplifiers to excite servomotors which stabilize the antenna in roll, pitch, and drift.

3.5.1.1.3 Timing Signals - A program to generate the timing signals required for scanning the antenna.

3.5.1.1.4 Synchronizer - A synchronizer to provide timing signals for radar system operation.

3.5.1.1.5 Dither Functions - A program to generate the dither functions to the transmitter.

3.5.1.1.6 Ancillary Circuits - Ancillary circuits necessary for system operation.

3.5.1.2 Form Factor - The power supply-programmer shall be contained within the following dimensions:

13.08 inch (33.22 cm) high

5.76 inch (14.63 cm) wide

22.10 inch (56.13 cm) deep

3.5.1.3 Weight - The weight of the power supply-programmer shall not exceed 42 pounds (19.05 kg).

3.5.1.4 Controls - The power supply programmer shall have the following controls:

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3.5.1.4.1 XMTR DISABLE - The XMTR DISABLE switch shall be a 3-position toggle switch which shall place the radar transmitter in operate, off, or dummy load positions.

3.5.1.4.2 BRAKE RELEASE - The BRAKE RELEASE switch shall be a momentary contact pushbutton switch to enable the antenna to rotate in roll, azimuth, and elevation with the radar turned off.

3.5.1.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
1J1	MS3113H14C5P	Aircraft input power
1J2	MS3120E22-55S	General purpose
1J3	MS3120E22-55SZ	Power and control signals
1J4	MS3120E24-61S	Power and control signals
1J5	MS3112E18-32S	Test signals
1J6	L22TE95SON2H49 MS3116, Type E	Power and control signals

3.5.1.6 Mounting - The power supply-programmer shall be hard mounted to the Forward Assembly Mount, MT-4043/APQ-126(V).

3.5.1.7 Short Protection - The power supply shall be protected against shorts.

3.5.2 Transmitter, Radar, T-1091/APQ-126(V) - The transmitter shall meet the following requirements:

3.5.2.1 Function - The radar transmitter shall generate and transmit a Ku-band pulse-modulated radio frequency signal and shall supply a detected sample of the transmitter pulse. It will also generate a master trigger for system timing.

3.5.2.2 Form Factor - The radar transmitter shall be contained within the following dimensions:

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5.63 inch (14.30 cm) high

18.28 inch (46.43 cm) wide

15.76 inch (40.03 cm) deep

3.5.2.3 Weight - The weight of the radar transmitter shall not exceed 42 pounds (19.05 kg).

3.5.2.4 Pressurization Switch - The pressurization switch shall continuously monitor the pressure inside the waveguide and shall disable the transmitter in the event that the waveguide pressure drops below 12.0 + 0.5 PSIA (3.37 kg/cm^{2}ABS). A visual fault detector shall indicate that the pressure switch has actuated.

3.5.2.5 Controls - The radar transmitter shall have no in-flight controls.

3.5.2.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
2J1	MS3124E14-19PX	Input power
2J2	DAGE 1-707-1	Modulator trigger
2J3	CANNON DAMA-15P-FO	Tuning servo signals and voltages
2J4	CANNON DBM-9W4PA156	Detected pulses and +20 volts dc routing
2J5	MS3114H8C-3P	Transmitter operate lines to pressure switch

3.5.2.7 Mounting - The radar transmitter shall be hard mounted to the Forward Assembly Mount, MT-4043/APQ-126(V).

3.5.3 Antenna-Receiver AS-2272/APQ-126(V) - The antenna-receiver shall meet the following requirements:

3.5.3.1 Function - The antenna-receiver shall stabilize and scan the radar beam in the required radiation pattern and shall process return radar signals to obtain sum, difference, and monopulse resolution improved (MRI) video. It shall also provide a test target for radar system performance monitoring.

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3.5.3.2 Form Factor - The antenna-receiver shall be contained within the following dimensions:

20.33 inch (51.64 cm) high

19.50 inch (49.53 cm) wide

22.84 inch (58.01 cm) deep

3.5.3.3 Weight - The weight of the antenna-receiver shall not exceed 44.5 pounds (20.18 kg).

3.5.3.4 Controls - The antenna-receiver test panel shall have the following controls:

3.5.3.4.1 LO, AFC - The LO AFC switch shall be a 3-position toggle switch which shall place the LO in the CTR, OFF, or BCN position.

3.5.3.4.2 LO, CTR - The LO CTR control shall be a potentiometer to adjust to LO center frequency. The potentiometer shaft shall be locked in place when not being adjusted.

3.5.3.4.3 LO, BCN - The LO BCN control shall be a potentiometer to adjust the LO center frequency in the BCN mode. The potentiometer shaft shall be locked in place when not being adjusted.

3.5.3.4.4 GAIN, SET - The SET GAIN control shall be a potentiometer to adjust the IF postamplifier gain in TF, TA, and the cross-scan modes. The potentiometer shaft shall be locked in place when not being adjusted.

3.5.3.4.5 GAIN, MAX - The MAX GAIN control shall be a potentiometer to adjust the IF postamplifier gain in BCN and AGR modes. The potentiometer shaft shall be locked in place when not being adjusted.

3.5.3.4.6 TEST SELECT - The TEST SELECT switch shall be a 9-position wafer switch to direct test signals to the Test Panel Meter.

3.5.3.4.7 PUSH-TO-READ - The PUSH-TO-READ button shall be a normally open pushbutton switch to operate the Test Panel Meter.

3.5.3.5 Electrical Connections - Connections to external circuits shall be provided as follows:

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Reference Designation	Receptacle Type	Function
3J1	MS3120E18-32P	Logic and control signals
3J2	M22T10XS2N-H2	Radar signals
3J3	DCMA-37P	Elevation, Servo and waveguide logic signals
3J4	DCMA-37P	Azimuth Servo

3.5.3.6 Mounting - The antenna-receiver shall be hard mounted to the Forward Assembly Mount, MT-4043/APQ-126(V).

3.5.4 Computer, Air Navigation, CP-954/APQ-126(V) - The air navigation computer shall meet the following requirements:

3.5.4.1 Function - The air navigation computer shall utilize the master trigger, MRI video, elevation scan information, velocity, set clearance information, altitude data and flight path angle to derive the proper commands for presentation on the attitude direction indicator and Head Up Display. The air navigation computer shall contain circuits to test the performance of the five functions given in 3.4.4.1.

3.5.4.2 Form Factor - The air navigation computer shall be contained within the following dimensions:

4.99 inch (12.67 cm) high

12.13 inch (30.81 cm) wide

8.88 inch (22.56 cm) deep

3.5.4.3 Weight - The weight of the air navigation computer shall not exceed 7.8 pounds (3.54 kg).

3.5.4.4 Controls - The air navigation computer shall contain no in-flight controls.

3.5.4.5 Electrical Connections - Connections to external circuits shall be provided as follows:

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Reference Designation	Receptacle Type	Function
4J1	MS3120E24-61P	Power and control signals
4J2	DAGE 1-707	Test oscillator pulse
4J3	DAGE 1-707	AFC BIT
4J4	DAGE 1-707	Template video
4J5	DAGE 1-707	Selected video
4J6	DAGE 1-707	MRI video
4J7	DAGE 1-707	Sum video
4J8	DAGE 1-707	Transmitter power sample
4J9	DAGE 1-707	Master Trigger

3.5.4.6 Mounting - The air navigation computer shall be hard mounted to the Forward Assembly Mount, MT-4043/APQ-126(V).

3.5.5 Control, Radar Set (Red Lighting) C-8251/APQ-126(V), Control, Radar Set (White Lighting) C-8252/APQ-126(V) - These Radar Set Controls shall meet the following requirements:

3.5.5.1 Function - The Radar Set Control shall provide selection of the radar mode, logic for system operation, frequency, antenna tilt, antenna polarization and the master power switch.

3.5.5.2 Form Factor - The Radar Set Control shall be contained within the following dimensions:

3.75 inch (9.53 cm) high

5.75 inch (14.61 cm) wide

6.80 inch (12.27 cm) deep

3.5.5.3 Weight - The weight of the radar set control shall not exceed 3.1 lbs (1.41 kg).

3.5.5.4 Content - The Radar Set Control shall contain the necessary switches, potentiometers, relays, and other components necessary for system operation.

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3.5.5.5 Controls - The Radar Set Control shall have the following controls:

3.5.5.5.1 POWER - The POWER switch shall be a 3-position toggle switch which shall place the radar in the OFF, STANDBY, or ON condition.

3.5.5.5.2 FREQ - The FREQ control shall be a thumbwheel rotary switch to select the transmitter center frequency.

3.5.5.5.3 POLAR - The POLAR switch shall be a 2-position toggle switch to select linear or circular antenna polarization.

3.5.5.5.4 ANT TILT - The ANT TILT control shall be a potentiometer to adjust the desired antenna tilt.

3.5.5.5.5 MODE - The MODE switch shall be a multiposition push button switch used to select the BCN, TA, GMP, GMS, or AGR mode of operation.

3.5.5.6 Electrical Connections - Connections to external circuit shall be provided as follows:

Reference Designation	Receptacle Type	Function
5J1	Bendix JPTOORE-24-61P	Power and logic

3.5.5.7 Mounting - The Radar Set Control shall be hard mounted to the airframe in the cockpit.

3.5.6 Control Radar Set (Instrument Panel Mount) C-8253/APQ-126(V) and Control Radar Set (Instrument Panel Mount) C-8255/APQ-126(V) - These Radar Set Controls shall meet the following requirements:

3.5.6.1 Function - The Radar Set Control shall provide the selection of terrain clearance and range set.

3.5.6.2 Form Factor - The Radar Set Control shall be contained within the following dimensions:

3.40 inch (8.64 cm) high

2.12 inch (5.38 cm) wide

4.70 inch (11.94 cm) deep

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3.5.6.3 Weight - The weight of the Radar Set Control shall not exceed 1 pound (.453 kg).

3.5.6.4 Content - The Radar Set Control shall contain the necessary switches and other components necessary for system operation. The Radar Set Control shall also contain the BCD counter for generating the manual range cursor in the ground mapping modes, and for generating in-range indication in AGR mode.

3.5.6.5 Controls - The Radar Set Control shall have the following controls:

3.5.6.5.1 TER CLR - The TER CLR control shall be a thumbwheel rotary switch to select the terrain clearance voltage in the terrain following mode.

3.5.6.5.2 RANGE SET - The RANGE SET control shall be composed of three thumbwheel rotary switches to select feet X 1000 in the AGR mode or NMI in the GMP/S modes.

3.5.6.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designator	Receptacle Type	Function	Radar Set Control
10J1	Hughes MRP0052P20BN000	Power and AGR Signals	C-8254/APQ-126(V)
10J1	Hughes MRK0052P61JN600	Power and AGR Signals	C-8255/APQ-126(V) OR C-8253/APQ-126(V)

3.5.6.7 Mounting - The Radar Set Control shall be hard mounted to the airframe in the cockpit.

3.5.7 Generator, Sweep, SG-811/APQ-126(V) - The sweep generator shall meet the following requirements.

3.5.7.1 Function - The sweep generator shall contain the circuits required to generate horizontal and vertical sweep waveforms, unblanking gates, and other pulse waveforms used to display radar and TV information on the indicator DVST. The sweep generator shall also contain the AGR ranging computer, fault isolation functions and other circuits as required for system operation.

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3.5.7.2 Form Factor - The sweep generator shall be contained within the following dimensions:

6.56 inch (16.66 cm) high

9.50 inch (24.13 cm) wide

22.00 inch (55.88) deep

3.5.7.3 Weight - The weight of the sweep generator shall not exceed 28 pounds (12.7 kg).

3.5.7.4 Controls - The sweep generator shall contain no in-flight controls.

3.5.7.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
6J1	Bendix JTP00RE-24-61P	Power and signals
6J2	Bendix JTP00RE-24-61PA	Power and signals
6J3	Deutsch CVC6063E-61-21P-1	Radar signals
6J4	Deutsch CVC6063E-61-21PW	Radar signals
6J5	Bendix JTP00RE-24-61PB	Power and signals
6J6	Bendix JTP00RE-24-61-PC	Power and signals
6J7	Bendix JTP00RE-24-61S	Test signals

3.5.7.6 Mounting - The sweep generator shall be hard mounted to the airframe behind the cockpit area.

3.5.8 Fault Locator, Radar Set, MX-8195/APQ-126(V) - The radar set fault locator shall meet the following requirements:

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3.5.8.1 Function - The fault locator shall provide a means of isolating a failure of the radar down to the LRU level. The fault locator shall also be capable of indicating failure of associated aircraft equipment.

3.5.8.2 Form Factor - The fault isolator shall be contained within the following dimensions:

3.00 inch (7.62 cm) high

4.50 inch (11.43 cm) wide

5.75 inch (14.61 cm) deep

3.5.8.3 Weight - The weight of the fault locator shall not exceed 1 pound(0.453 kg).

3.5.8.4 Content - The fault locator shall contain the switching circuitry necessary to connect the fault isolation circuitry contained in the sweep generator LRU to the specific LRU selected by the fault locator switch. The fault locator shall contain a SAFE lamp indicating the condition of the LRU under test.

3.5.8.5 Controls - The fault locator shall have the following control:

a. FAULT LOCATION - The FAULT LOCATION switch shall be a 4-pole, 12-position, 4-wafer switch used to select either OFF, BIT, ACFT, P/S, XMTR, CMPTR, A/R, 2, 3, SW GEN, or SCAN INTER fault isolation function.

3.5.8.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
7J1	Bendix JTPOORE-18-32P	Radar signals

3.5.8.7 Mounting - The fault locator shall be hard mounted to the airframe in the cockpit area.

3.5.9 Indicator, Multiple Air Navigation (Red Lighting), IP-940/APQ-126(V) and Indicator, Multiple Air Navigation (White Lighting), IP-952/APQ-126(V) - The indicator shall meet the following requirements:

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3.5.9.1 Function - The indicator shall display radar, TV, SIDS, BCN and AGR information on a 5-inch (12.7 cm), high-resolution DVST. For TV display purposes, the indicator shall have a minimum of 4 inches (10.16 cm) (diagonal) of usable display, a minimum resolution of 400 lines for a 500-line output, and six contrasting shades of gray under cockpit ambient lighting conditions. The tube shall have a minimum resolution of 100 shrinking raster lines per inch with 1000 foot-lamberts brightness at the face of the tube. The indicator shall have an engraved transparent scale which shall be etched on the face of the indicator and shall define the radar display scales and references. Lighting shall be provided for the transparent scale. A separately and independently lighted scale shall be provided for the SIDS mode such that, when the SIDS mode is selected, only the SIDS scale shall be lighted. The SIDS scale shall not be lighted during other mode selections.

3.5.9.2 Form Factor - The indicator shall be contained within the following dimensions:

5.75 inch (14.61 cm) high

6.99 inch (17.75 cm) wide

19.46 inch (49.42 cm) deep

3.5.9.3 Weight - The weight of the indicator shall not exceed 21.1 pounds (9.57 kg).

3.5.9.4 Controls - The indicator shall have the following controls.

3.5.9.4.1 TV/SHRIKE-OVERRIDE - The TV/SHRIKE-OVERRIDE switch shall be a DPDT switch to override TV and SHRIKE available signal.

3.5.9.4.2 CONTRAST - The CONTRAST control shall be a potentiometer to control the TV video amplitude.

3.5.9.4.3 VID GAIN - The VID GAIN control shall be a potentiometer to control the radar video amplitude.

3.5.9.4.4 SCALE BRT - The SCALE BRT control shall be a potentiometer to control the transparent etched scale brightness from lamp cutoff to the maximum attainable brightness from the lamps.

3.5.9.4.5 RCVR GAIN - The RCVR GAIN control shall be a potentiometer to control the receiver gain from cutoff to maximum gain in the map modes only.

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3.5.9.4.6 BRIL - The BRIL control shall be a potentiometer to control DVST brilliance from DVST cutoff to maximum attainable brightness.

3.5.9.4.7 STOR TIME - The STOR TIME control shall be a potentiometer to control DVST storage time from zero to a maximum of 3 seconds.

3.5.9.4.8 CURSOR - The CURSOR control shall be a potentiometer to vary both the range and azimuth cursor brilliance.

3.5.9.5 Lamp Indicators - The indicator will provide for the following lamp indications.

Fail light

TV light

SHRIKE light

Range lamps - shall have range indicators for 5, 10, 20, 40, 80 miles.

3.5.9.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
8J1	Cannon DPKB-71C15-PC3	Power and signals

3.5.9.7 Mounting - The indicator shall be hard mounted to the airframe in the cockpit.

3.5.10 Base, Shock Mount, Electrical Equipment, MT-4043/APQ-126(V) - The shock mount base shall meet the following requirements.

3.5.10.1 Function - The shock mount base shall consist of a machined magnesium casting, left and right shock isolators, an adjustable boresight plate, roll stabilization gimbal, and two connector brackets. The base shall attach to the airframe by four cast lugs located on the left and right shock isolators.

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3.5.10.2 Form Factor - The shock mount base shall be contained within the following dimensions:

14.932 inch (37.93 cm) high

24.32 inch (61.77 cm) wide

7.95 inch (20.19 cm) deep

3.5.10.3 Weight - The weight of the shock mount base shall not exceed 36 pounds (16.33 kg).

3.5.10.4 Content - The shock mount base shall contain the roll drive assembly blower fan, and interconnect cable. The antenna-receiver shall mount to the roll plate which mounts to the adjustable boresight plate at the front of the shock mount base. The radar transmitter shall be bolted to the top machined surface of the shock mount base. The air navigation computer shall be bolted to the underside of the shock mount base. Provisions shall be made to secure seven connectors on the shock mount base. Provisions shall be made for securing the wiring harness to the shock mount base.

3.5.10.5 Controls - The shock mount base shall contain no in-flight control.

3.5.10.6 Electrical Connections - Connections to external circuits shall be provided as follows:

NOTE: Harness has 3 wires with eyelet ends (no plug) for blower power.

Reference Designation	Receptacle Type	Function
1P2	MS3126F22-55P	Power and signals
1P3	MS3126F22-55PZ	Power and signals
1P4	MS3126E24-61P	Power and signals
1P6	L22TE95P6N2H49	Power and signals
2P1	PTO8SE-14-19SX(SR)	Power
2P2	5347-1 (Polarized E)	Modulator trigger
2P3	DAMA-15S-FO	Power, position analog, and CF tune

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Reference Designation	Receptacle Type	Function
2P4	DBM-9W4S-A156	Power and AGR trigger
2P5	MS3116P8-35	Control signals
3-Wire	Swaged Eyelets	Blower power
3P1	MS3126F18-32S	Power and signals
3P2	M22TAC10XP6N-H2	Video and gating
3P3	DCMA-37S-FO	Power
3P4	DCMA-37S-FO	Power and signals
4P1	MS3126F24-61S	Power and signals
4P2	5347-1 (Polarized E)	Test oscillator trigger
4P3	5347-1 (Polarized E)	AFC BIT
4P4	5347-1 (Polarized E)	Template video
4P5	5347-1 (Polarized E)	Selected video
4P6	5347-1 (Polarized E)	MRI video
4P7	5347-1 (Polarized E)	SUM video
4P8	5347-1 (Polarized E)	XTMR power sample
4P9	5347-1 (Polarized E)	Master trigger
12P1	DBMA-25S-FO	Power and roll generator
W1J1	MS27497E18A32P1	Power and controls
W1J2	M22T10XS2N-H2	Video and master trigger
W1J3	MS27497E24A-61S1	Power and controls
W1J4	MS27497E24A-61SA	Power and controls
W1J5	MS27497E24A-61S1	Power and controls
W1J6	M22T10X52N-H2	Power, controls, and tests

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3.5.10.7 Mounting - The shock mount base shall be shock mounted to the air frame in the radome area by four cast lugs located on the left and right shock isolators.

3.5.11 Fan, Centrifugal HD-821A - The centrifugal fan shall meet the following requirements.

3.5.11.1 Function - The centrifugal fan shall provide cooling air to the power supply, transmitter and magnetron in the radar transmitter sector during system operation.

3.5.11.2 Form Factor - The centrifugal fan shall be contained within the following dimensions:

8.34 inch (21.18 cm) high

15.07 inch (38.28 cm) wide

4.92 inch (12.50 cm) deep

3.5.11.3 Weight - The weight of the centrifugal fan shall not exceed 4.9 pounds (2.22 kg).

3.5.11.4 Content - The centrifugal fan shall be an electrically operated fan and shall include an air duct.

3.5.11.5 Controls - The centrifugal fan shall contain no in-flight controls.

3.5.11.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
14J1	Bendix JTP00RE-10-08P	Power

3.5.11.7 Mounting - The centrifugal fan shall be shock mounted to the airframe in the radome area.

3.5.12 Convertor Signal Data CV-3368/APQ-126(V) - The signal data convertor shall meet the following requirements:

3.5.12.1 Function - The signal data convertor shall contain the necessary circuitry for radar video processing, air-to-ground ranging computations and data transmission, cursor generation, composite video generation and other waveforms and discretes necessary to provide display and radar system required operation.

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3.5.12.2 Form Factor - The signal data convertor shall be contained within the following dimensions:

6.56 inch (16.66 cm) high

9.53 inch (24.21 cm) wide

22.08 inch (56.08 cm) deep

3.5.12.3 Weight - The weight of the signal data convertor shall not exceed 28 pounds (12.71 kg).

3.5.12.4 Controls - The signal data convertor shall contain no in-flight controls.

3.5.12.5 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
6J1	Bendix JTP00RE-24-61P	Power and signals
6J2	Bendix JTP00RE-24-61PA	Power and signals
6J3	Deutsch CVC6063E-61-21P-1	Radar signals
6J4	Deutsch CVC6063E-61-21PW	Radar signals
6J5	Bendix JTP00RE-24-61-PB	Power and signals
6J6	Bendix JTP00RE-24-61-PC	Power and signals
6J7	Bendix 33506-2	Test signal

3.5.12.6 Mounting - The signal data convertor shall be hard mounted to the airframe behind the cockpit area.

3.5.13 Indicator, Intratarget Data, FWD (Red Lighting), IP-1256/APQ-126(V) - The indicator shall meet the following requirements.

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3.5.13.1 Function - The indicator shall display radar, TV, SIDS, BCN and AGR information on a 5-inch (12.7 cm), high-resolution CRT. For TV display purposes, the indicator shall have a minimum of 4 inches (10.16 cm) (diagonal) of usable display, a minimum resolution of 600 lines for a 700-line input, and three contrasting shades of gray under cockpit ambient lighting conditions of 10,000 foot-lamberts. The tube shall have a minimum resolution of 185 shrinking raster lines per inch with 1000 foot-lamberts brightness at the face of the tube. The indicator shall have an engraved transparent scale which shall be etched on the face of the indicator and shall define the radar display scales and references. Lighting shall be provided for the transparent scale. A separately and independently lighted scale shall be provided for the SIDS mode such that, when the SIDS mode is selected, only the SIDS scale shall be lighted. The SIDS scale shall not be lighted during other mode selections.

3.5.13.2 Form Factor - The indicator shall be contained within the following dimensions:

5.81 inch (14.76 cm) high

7.03 inch (17.86 cm) wide

19.51 inch (49.56 cm) deep

3.5.13.3 Weight - The weight of the indicator shall not exceed 21.9 pounds (9.94 kg).

3.5.13.4 Controls - The indicator shall have the following controls.

3.5.13.4.1 TV/SHRIKE-OVERRIDE - The TV/SHRIKE-OVERRIDE switch shall be DPDT switch to override TV and SHRIKE available signal.

3.5.13.4.2 CONTRAST - The CONTRAST control shall be a potentiometer to control the display video amplitude.

3.5.13.4.3 EXPAND/NORMAL/HOLD - The Expand/Normal/Hold switch shall be a toggle switch to control display modes for the respective mode selections.

3.5.13.4.4 SCALE BRT - The SCALE BRT control shall be a potentiometer to control the transparent etched scale brightness from lamp cutoff to the maximum attainable brightness from the lamps.

3.5.13.4.5 RCVR GAIN - The RCVR GAIN control shall be a potentiometer to control the receiver gain from cutoff to maximum gain in the map modes only.

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3.5.13.4.6 BRIL - The BRIL control shall be a potentiometer to control CRT brilliance from cutoff to maximum attainable brightness.

3.5.13.4.7 CURSOR - The CURSOR control shall be a potentiometer to vary both the range and azimuth cursor brilliance.

3.5.13.5 Lamp Indicators - The indicator will provide for the following lamp indications.

Fail light

TV light

Hold light

SHRIKE light

Range lamps - shall have range indicators for 5, 10, 20, 40, 80 miles.

3.5.13.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
8J1	Cannon DPKB-71C15-PC3	Power and signals

3.5.13.7 Mounting - The indicator shall be hard mounted to the airframe in the cockpit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified, the contractor is responsible for the performance of all test requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the tests set forth in the specification where such tests are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

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a. Preproduction (First Article) Tests

b. Acceptance Tests

4.2 Preproduction Tests - Preproduction tests are tests conducted on samples which are representative in every way of production equipment to determine conformance of the equipment with the requirements of this specification.

4.2.1 Sampling - Preproduction tests shall be performed on equipment representative of the production item in every respect. The equipment shall successfully pass all tests of the procuring activity approved Preproduction Test Program.

4.2.2 Scope of Tests - Preproduction tests shall be devised and conducted by the contractor or an approved testing agency. Tests shall include all tests deemed necessary to determine that the equipment satisfactorily meets all the requirements of the specification. Preproduction tests shall include successful completion of environmental tests in accordance with Specification MIL-T-5422 to prove compliance with the requirements specified in paragraph 3.3.10 and interference tests in accordance with Specification MIL-I-6181 except as modified herein. Environmental tests required for specific units are listed in the following paragraphs.

4.2.2.1 Temperature-Altitude -

4.2.2.1.1 -54°C to +71°C from sea level to 50,000 ft. (15,240 m) as bounded by Figure 6, Sheet 2, Curve A of MIL-E-5400 for Class II equipment applicable for components other than the forward assembly, indicator, and set control.

4.2.2.1.2 -54°C to +55°C as bounded by Figure 6, Sheet 1, Curve A of MIL-E-5400 for Class I equipment applicable for the Forward Assembly Group, Indicator and Radar Set Control.

4.2.2.2 Humidity - MIL-T-5422 paragraph 4.4.

4.2.2.3 Vibration -

4.2.2.3.1 Forward Assembly, Centrifugal Fan and Divergent Duct - The Forward Assembly and centrifugal fan with divergent duct shall be tested in accordance with MIL-T-5422 except that Figure 1 and Table IV of this specification shall apply.

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4.2.2.3.2 Sweep Generator - The sweep generator shall be tested in accordance with MIL-T-5422 except that Figure 2 of this specification shall apply.

4.2.2.3.3 Fault Locator and Radar Set Control - The fault locator and radar set control shall be tested in accordance with MIL-T-5422 except Figure 3 and Table IV of this specification shall apply.

4.2.2.3.4 Indicator - The indicator shall be tested in accordance with MIL-T-5422 except that Figure 4 of this specification shall apply.

4.2.2.4 Shock - MIL-T-5422, paragraph 4.3 shall apply, with additional shock loads as specified in paragraph 3.3.10.4 of this specification.

4.2.2.5 Sand and Dust - MIL-T-5422, paragraph 4.7.

4.2.2.6 Fungus - MIL-T-5422, paragraph 4.8.

4.2.2.7 Salt Atmosphere - MIL-T-5422, paragraph 4.5.

4.2.2.8 Explosive Conditions - MIL-T-5422, paragraph 4.6.

4.2.2.9 Electromagnetic Interference Tests - The system shall comply with the applicable requirements of MIL-I-6181 with the following exceptions:

a. Narrow band (CW) conducted interference limits using stabilization networks shall be as shown in Figure 5 of this specification in lieu of MIL-I-6181D, Figure 7.

b. Broad band and pulsed CW conducted interference limits using stabilization networks shall be as shown in Figure 6 of this specification in lieu of MIL-I-6181D, Figure 8.

c. Narrow band (CW) conducted interference limits using a current probe shall be as shown in Figure 7 of this specification in lieu of MIL-I-6181D, Figure 9.

d. Broad band and pulsed CW conducted interference limits using a current probe shall be as shown in Figure 8 of this specification in lieu of MIL-I-6181D, Figure 10.

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e. Narrow band (CW) radiated interference limits shall be as shown in Figure 9 of this specification in lieu of MIL-I-6181D, Figure 14.

f. Broad band and pulsed CW radiated interference limits shall be as shown in Figure 10 of this specification in lieu of MIL-I-6181D, Figure 16.

g. The audio frequency conducted susceptibility requirements of paragraph 4.3.4.1.2 of MIL-I-6181D shall apply except that the open circuit rms voltage level input shall be adjusted to the following levels at the following corresponding frequencies:

2.7 volts at 335 Hz

2.4 volts at 500 Hz

2.1 volts at 592 Hz

1.5 volts at 830 Hz

4.2.2.10 Acoustic Noise Tests - MIL-STD-810A (USAF), Method 515, Grade A. System shall be operating during test.

4.2.3 Equipment Changes - After completion of preproduction tests, the contractor shall make no changes which may affect weight, performance, installation, interchangeability and interchangeability of maintenance parts or assemblies as specified in MIL-STD-480, unless approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be as construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests

b. Reliability Assurance Tests

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4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability.

4.3.1.1 Scope of Tests - Each LRU of a radar set submitted for acceptance under contract shall be tested both individually and as part of the assembled radar set. After proof of functional compliance in the system alignment test, each assembled system shall be subjected to a final acceptance test which shall require successful completion without failure or adjustment. Each LRU submitted for acceptance as a spare unit shall require successful completion of a final acceptance test without failure or adjustment and a suitable burn-in time as specified by the contractor and approved by the procuring activity.

4.3.1.2 Each LRU of a radar set submitted for acceptance under contract shall be tested both individually and as part of the assembled radar set. After proof of functional compliance in the system alignment test, each assembled system shall be subjected to a final acceptance test and burn-in per paragraph 4.3.1.5. Each LRU submitted for acceptance as a spare unit shall be subjected to a final acceptance test which shall require successful completion and a suitable burn-in time as specified by paragraphs 4.3.1.6 and 4.3.1.7.

4.3.1.3 The procuring activity may reserve the right to require modification of the contractor's test program if it does not adequately demonstrate compliance with the procurement specification. The procuring activity may also reserve the right to require a demonstration that any LRU of the radar set, selected at random after testing by its individual test procedure, can be interchanged with another identical LRU in an assembled radar set and the set harmonized with the adjustments and procedures specified by the contractor as required in paragraph 3.3.7 for organizational maintenance.

4.3.1.4 Upon completion of individual tests, forward assembly groups shall remain assembled as a group in order to maintain pressurization integrity, etc. unless the procurement is limited to an individual LRU which would obviate this requirement.

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4.3.1.5 Each system (less Fault Locator) shall be subjected to a thermal and vibration profile as shown in Figure 11 until a minimum of 96 operational hours has been accumulated on the Elapsed Time Indicator from the initial reading at the first burn-in cycle. The last cycle must be failure free.

4.3.1.6 Each LRU of the forward assembly group (except Centrifugal Fan and Shock Mount Base) to be supplied as spares shall be subjected to a thermal and vibration profile as shown in Figure 12. A minimum of fifty (50) operating hours shall be accumulated.

4.3.1.7 The Sweep Generator, Convertor and Indicator of the cockpit group to be supplied as spares shall be subjected to a thermal and vibration profile as shown in Figure 13. A minimum of fifty (50) operating hours shall be accumulated on each LRU.

4.4 Reliability Testing - Testing shall be performed to demonstrate a specified MTBF of 106 hours as per MIL-STD-781 except as modified herein. The test shall be conducted on a sample of 5 production radar systems; test duration shall be 848, 954, or 1060 hours total equipment operating hours. The following factors further define the reliability test requirements.

4.4.1 Test Level -

4.4.1.1 Temperature - -54°C ± 2°C to +57°C ± 2°C

4.4.1.2 Cooling Air -

			Sweep Generator & Converter	Forward Assembly	Set Control Indicator Fault Locator
Max. Amb. Temp.	Flow Rate #(kg)/Min			7.5 (3.4)	0
	Cooling Air Temp °C			30	-
Min. Amb. Temp.	Flow Rate #(kg)/Min			7.5 (3.4)	0
	Cooling Air Temp °C	Ambient		30	-

Rate of change of cooling air temperature shall be approximately linear between the minimum and maximum ambient temperatures. For the forward assembly, the

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cooling air shall be dumped into the plenum to simulate the actual aircraft installation as defined in paragraph 3.3.13.1.1. Cooling air for the sweep generator shall be ducted to a shroud in accordance with paragraph 3.3.13.1.2.

4.4.1.3 Temperature Cycling - In accordance with paragraph 5.2.3 of MIL-STD-781, except high temperature operation (Period B of Figure 1) shall be two hours per cycle and Period A shall not be greater than 1.0 hours.

4.4.1.4 Vibration - +2.2 g's ± 10 percent peak vibration or that value specified in paragraph 3.3.10.1, whichever is less, at any non-resonant frequency between 20 and 60 Hz measured at the mounting point on the equipment. Vibration shall be applied continuously when the equipment is operating throughout the test duration.

4.4.1.5 Equipment ON-OFF Cycling - Equipment off during cooling cycle and on during heating cycle in accordance with paragraph 5.2.1 of MIL-STD-781.

4.4.2 Test Requirements - The requirements of paragraph 5.0 of MIL-STD-781 apply except where modified below or in the contractor's approved test plan:

4.4.2.1 Equipment operation shall be checked at the end of each cycle by use of BITE; performance tests shall be performed at intervals to be defined by the contractor and approved by the procuring activity. Failures discovered during the performance tests shall be considered to have occured at the midpoint of the test interval for purposes of making an accept-reject decision.

4.4.2.2 Failure is defined as unscheduled item removal, repair, or adjustment, except those adjustments which are accessible in flight shall not be counted. Failures shall be as determined by observation, equipment checks, or performance tests and shall include failure to meet the performance requirements specified herein. Relevant failures shall be defined by the contractor and submitted for approval by the procuring activity with the test plan. During the test, the procuring activity reserves the right of final approval of failure classification.

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4.4.2.3 Preventive maintenance type and frequency shall be defined by the contractor in the test plan and subject to approval by the procuring activity. Such maintenance shall not exceed the allowable during operational usage. Parts which have been discovered to have deteriorated but are still within specified performance limits shall not be replaced during the reliability tests.

4.4.2.4 Reliability test units shall be refurbished, as necessary, at the completion of the test and submitted for acceptance. Units shall be clearly identified as test samples until the refurbishing process has been completed.

4.4.2.5 Any corrective action required as a result of reliability tests shall be the responsibility of the contractor in accordance with the contract. The procuring activity may reserve the right to witness any portion or all of the tests required herein.

4.5 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract.

4.5.1 Preproduction Tests - Submittals per 4.2.

4.5.2 Acceptance Tests - Submittals per 4.3.

4.6 Reconditioning of Tested Equipment -

4.6.1 Preproduction Test Samples - The approved preproduction test samples shall be retained by the contractor. The preproduction sample shall not be considered as deliverable equipment under the contract.

4.6.2 Reliability Test Samples - Equipment subjected to reliability tests shall be reconditioned as required by the contractor and submitted for acceptance. Test samples shall be clearly identified as such until the time that the reconditioning is complete.

4.7 Rejection and Retest -

4.7.1 Preproduction Tests - If the sample production item(s) fails to pass any preproduction test requirements, that particular preproduction test shall be stopped and the necessary corrective action

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taken. Full particulars of the failure, and corrective action taken, shall be reported to the procuring activity for approval at the time the action has been effected. The retesting necessary to demonstrate the environmental suitability of corrective fixes shall be as mutually agreed upon by the procuring activity and the contractor. Production drawings, schematics, etc., shall reflect all agreed upon changes prior to production and delivery of the equipment.

4.7.2 Acceptance Tests - Any equipment failing to pass an acceptance test shall be rejected and returned to the contractor.

4.7.3 Individual Tests - Equipment failing any individual test shall be rejected and may be repaired. When repaired items are submitted for acceptance, full particulars concerning failure and repairs shall be submitted with the equipment.

4.7.4 Reliability Assurance Tests - Test criteria shall be in accordance with the contractor's test plan as approved by the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing - The equipment shall be prepared for delivery in accordance with Specification MIL-STD-794, unit packaging to be level C and exterior packaging level C unless otherwise specified by the procuring activity. Forward Assembly Groups shall be packaged as an assembled group, without having been disassembled after individual system test, except where the procurement of an individual LRU obviates this requirement.

5.2 Marking for Shipment - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129 and shall include the following information:

Part name

Contractor's name and trademark

Contractor's address

Month and year of manufacture

Data of shipment

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6. NOTES

6.1 Intended Use - This radar set provides terrain following, terrain avoidance, ground mapping, air-to-ground ranging, TV display, SIDS display, and beacon mode for a forward-looking radar system. Combined terrain following/terrain avoidance, terrain following/ground mapping shall also be provided. When appropriately configured with Convertor (CV-3368/APQ-126(V)) and Indicator (IP-1256/APQ-126(V)), the radar set shall provide FLIR, Expanded Map and Frozen Map displays.

6.2 Ordering Data - The procuring activity should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification

b. Selection of applicable levels of packaging and packing (see 5.1).

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this radar set. Where a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and

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processes other than those specified in Specification MIL-E-5400 for Class 2, Category III equipment, it is desired their use be investigated. When such advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete support information.

6.5 Changes from Previous Issue - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.6 Associated Equipment -

Equipment	Part No.	Specification No.
Altimeter, Radar	AN/APN-141	MIL-A-23887
Inertial Measurement Set	AN/ASN-90	VAD 204-16-70
Indicator, Attitude Director	ID-1329	MIL-I-23524
Missile, Walleye	YASM-62A	WS-3180A
Three-Axis Rate Gyro Package	215-37128	VAD 204-15-24
Mixer, HUD Video		VAD 204-16-290E
Navigation/Weapon Delivery Digital Computer	AN/ASN-91(V) or AN/ASN-91A	MIL-N-85016
Computer, Air Data	CP-828/A	VAD 204-15-26
Beacon, Rendezvous	AN/APN-134	Boeing D10-60306
Convertor, Signal Conditioner	CV-2529A/APG	AS-1720/XWS-6517
Head-Up Display	AN/AVQ-7	VAD 204-16-19
Transformer	CVC 6070-5	MIL-T-27
Transformer	CVC 6070-6	MIL-T-27
Switch		MS-25089-3
Switch		MS-25089-3g

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6.7 This specification is under the cognizance of AIR-53332.

Preparing Activity:

NAVY - AS

Project No. 5841-NOO3

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Table I. Switching Requirements.

Main Power Switch	Master Function Discrete	Radar Mode Switch	Radar Mode	Radar Display	Radar Selection	Polarization	Frequency Selectable	Cursors	Manual Tilt
OFF	NA	NA	OFF	NA	NA	NA	NA	NA	NA
STBY	NA	BCN	STBY	None	NA	NA	NA	NA	NA
		TA	STBY	None	NA	NA	NA	NA	NA
		GMP	STBY	None	NA	NA	NA	NA	NA
		GMS	STBY	None	NA	NA	NA	NA	NA
		AGR	STBY	None	NA	NA	NA	NA	NA
OPR	None	BCN	BCN	$\pm$ 45°PPI	5,10,20,40,80	CIRC ONLY	NO	None	YES
		TA	TA	$\pm$ 45°PPI	5,10	SELECT	YES	None	NA
		GMP	GMP	$\pm$ 45°PPI	5,10,20,40,80	SELECT	YES	MAN RANGE	YES
		GMS	GMS	$\pm$ 45°PPI	5,10,20,40,80	LIN ONLY	YES	MAN RANGE	YES
		AGR	AGR	AGR	10 Only	LIN ONLY	NO	None	YES
	TFO	BCN	TF	E^2	10 Only	SELECT	YES	None	NA
		TA	CSTA	+ 20°PPI	5,10	SELECT	YES	None	NA
		GMP	CSGMP	$\pm$ 20°PPI	5,10	SELECT	YES	MAN RANGE	YES
		GMS	TF	E^2	10 Only	SELECT	YES	None	NA
		AGR	TF	E^2	10 Only	SELECT	YES	None	NA
	NORMAL ATTACK	BCN	AGR	AGR	10 Only	LIN ONLY	NO	None	NO
		TA	AGR	AGR	10 Only	LIN ONLY	NO	None	NO
		GMP	AGR	AGR	10 Only	LIN ONLY	NO	None	NO
		GMS	AGR	AGR	10 Only	LIN ONLY	NO	None	NO
		AGR	AGR	AGR	10 Only	LIN ONLY	NO	None	NO

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Table I. (Continued)

Main Power Switch	Master Function Discrete	Radar Mode Switch	Radar Mode	Radar Display	Radar Selection	Polarization	Frequency Selectable	Cursors	Manual Tilt
OPR	RADAR BOMB	BCN	GMP	± 45°PPI	5,10,20,40,80	SELECT	YES	Driven Range & Azimuth	YES
		TA	GMP	± 45°PPI	5,10,20,40,80	SELECT	YES	Driven Range & Azimuth	YES
		GMP	GMP	± 45°PPI	5,10,20,40,80	SELECT	YES	Driven Range & Azimuth	YES
		GMS	GMS	± 45°PPI	5,10,20,40,80	LIN ONLY	YES	Driven Range & Azimuth	YES
		AGR	GMP	± 45°PPI	5,10,20,40,80	SELECT	YES	Driven Range & Azimuth	YES

NOTES:

1. TV available will override any display
2. SIDS available will override any display
3. OVERRIDE switch will return display to previous selected display
4. Antenna is roll stabilized in all modes except STBY
5. Antenna is stabilized in Pitch and Yaw in all modes except STBY and AGR
6. Antenna is slaved in azimuth and elevation in AGR when so commanded by the computer. If no slaving command is received, azimuth is zero (0) degrees and elevation is controlled by the tilt control about the aircraft boresight line.

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Table II. Radar Output Interface.

FROM	SIGNAL FUNCTION	SCALE FACTOR	RANGE	TOLERANCE	TO
Radar	TF Fail Signal	Discrete	+28V		Head-Up Display
Radar	Radar Range #1	Digital	60K to 1K ft.	±26 ft or ±25%	Computer
Radar	Blanking Pulse	Pulse 5.4 ± 4 us	+15V ± 3V		Central Blanking Unit
Radar	Barom. Alt. Pot. Excitation Pot. #2	Constant	+30.5V	±0.1V	Air Data Computer
Radar	Flight Command Warning	Discrete - 245 microamps into 1000 load on ADI. The HUD warning symbology load shall not be less than 100K.			
Radar	Climb/Dive Commands	Analog - 2.2 Ma (ig) into 1K for 7/8° defl. on ADI. - 4.47 for full climb. HUD loading not to be less than 100K.			
Radar	TF Mode	Discrete	+28V		HMS
Radar	Set Range Readout	Discrete	+28V		Head-Up Display
Radar	3-phase Transformer Excitation	Constant	115 VAC @ 400 Hz		External Reference Transformers
Radar	Radar Alt. Fail	Discrete	Ground		Aircraft Switching Relay

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Table III. Radar Input Interface.

FROM	SIGNAL FUNCTION	SCALE FACTOR	RANGE	TOLERANCE	TO
IMS	Roll Signal	Syn. 206 mV/deg.	$\pm 180^{0}$ Cont.	$\pm 0.25^{0}$	Radar
IMS	Pitch Signal	Syn. 206 mV/deg.	$\pm 180^{0}$ Cont.	$\pm 0.25^{0}$	Radar
IMS	IMS Fail	Discrete	+28V		Radar
Computer	Antenna Azimuth Slave	Syn. 206 mV/deg.	± 40 deg.	$\pm .5^{0}$	Radar
Computer	Antenna Elevation Salve	Syn. 206 mV/deg.	± 20 deg.	$\pm 0.3^{0}$	Radar
Computer	Antenna Slave Discrete	Discrete	+28V		Radar
Computer	Drift Angle	Syn. 206 mV/deg.	$\pm 1^{0}$	$\pm 0.25^{0}$	Radar
Computer	Ground Velocity	Analog .02 V/kt	0 - 16V	$\pm 1\%$	Radar
Computer	Flight Path Angle	Analog 0.6 V/deg.	$\pm 30^{0}$	$\pm 0.25^{0}$	Radar
Computer	Computer Fail	Discrete	+28V Safe, 0V Fail		Radar
Computer	Range Word Test	Discrete	+28VDC		Radar
Computer	AGR Test	Discrete	+28V		Radar
AFCS	Yaw Rate	Analog 167 mV/deg/sec	± 30 deg/sec		Radar
Weapons Management	TV Available	Discrete	Ground		Radar
Weapons Management	TV Video	Refer to WS-3180A			Radar
Weapons Management	Snrike Selected	Discrete	Ground		Radar

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Table III. (Continued)

FROM	SIGNAL FUNCTION	SCALE FACTOR	RANGE	TOLERANCE	TO
(SIDS Box)	SIDS Vert/Horiz Deflection	0.4 volts/deg.	$\pm 25^{\circ}$		Radar
(SIDS Box)	SIDS Write Command	Discrete	+10 ± 2 VDC		Radar
ADC	Barometer Altitude	Analog 0.5 mV/ft	-1000 + 50,000 ft.	±1%	Radar
Radar Altimeter	Radar Altimeter	Analog 5 mV/ft	0 - 5,000 ft.	±7 ft or ±7%	Radar
Radar Altimeter	Radar Altitude Fail	Discrete	0V Fail, +5V Safe	±.5V	Radar
Master Function Switch	TF Mode	Discrete	+28V		Radar
Range Select Switch	Range Select	Switch Closure			Radar
Target Reject	Target Rejection	Switch Closure			Radar
External Reference Transformer	AC Power	Constant	26 VAC	ØC - 90 deg.	Radar
"	"	Constant	26 VAC	ØC	Radar
"	"	Constant	11.8 VAC	ØC	Radar
"	"	Constant	28.6 VAC	ØC	Radar
Computer	Driven Cursor Enable	Discrete	+28V		Radar
Computer	Cursor Position Data,AZ-sin	Digital	$\pm 45^{\circ}$		Radar
Computer	Cursor Position Data,AZ-cos	Digital	$\pm 45^{\circ}$		Radar
Computer	Cursor Position Data,Range	Digital	19.9 n. miles		Radar

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TABLE IV.

VIBRATION SCHEDULE

I. Resonance Survey:

Cycle from 5 Hz to 60 Hz at a nominal acceleration level such as 0.5 g. Determine the resonant frequencies by means of audible or visible observation and or vibration measurement devices.

II. Resonance Vibration: (5 Hz to 60 Hz)

No. of Res.	Dwell at each Resonance	Log Sweep 5-60 Hz
0	---	90 min.
1	30 min.	60 min.
2	30 min.	30 min.
3 1/	30 min.	---

1/ If more than 3 resonances exist then test the three major resonances.

III. Sweep Testing: (60 Hz to 2000 Hz)

Sweep testing will be done with logarithmic variation of frequency with time.

Total test time each direction	Sweep 60 to 2000 Hz	then sweep 2000 to 60 Hz
4 hrs.	15 min. up	15 min. down

IV. Total test - 16.5 hours.

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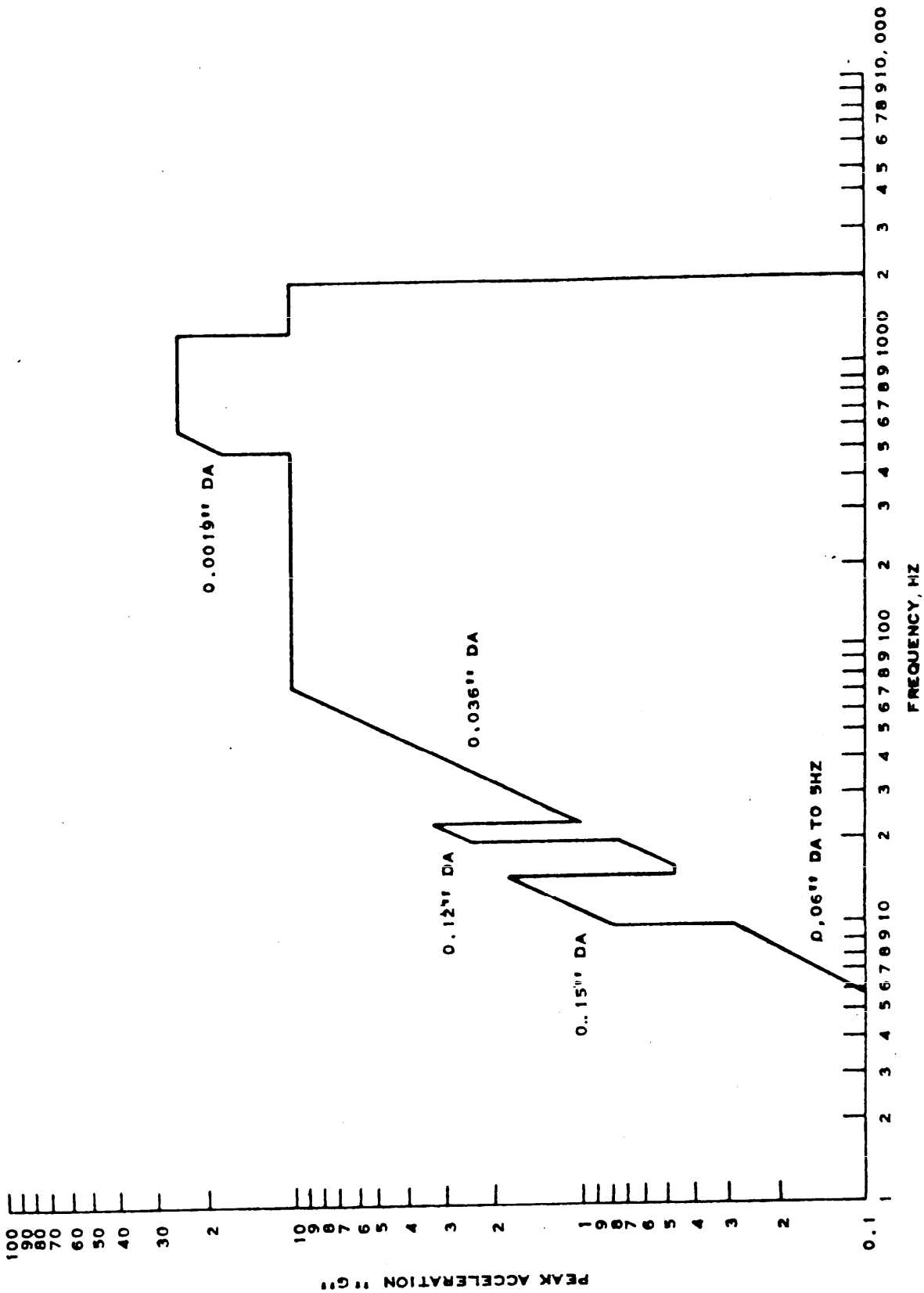


Figure 1. Vibration Curve, Forward Assembly and Centrifugal Fan with Divergent Duct.

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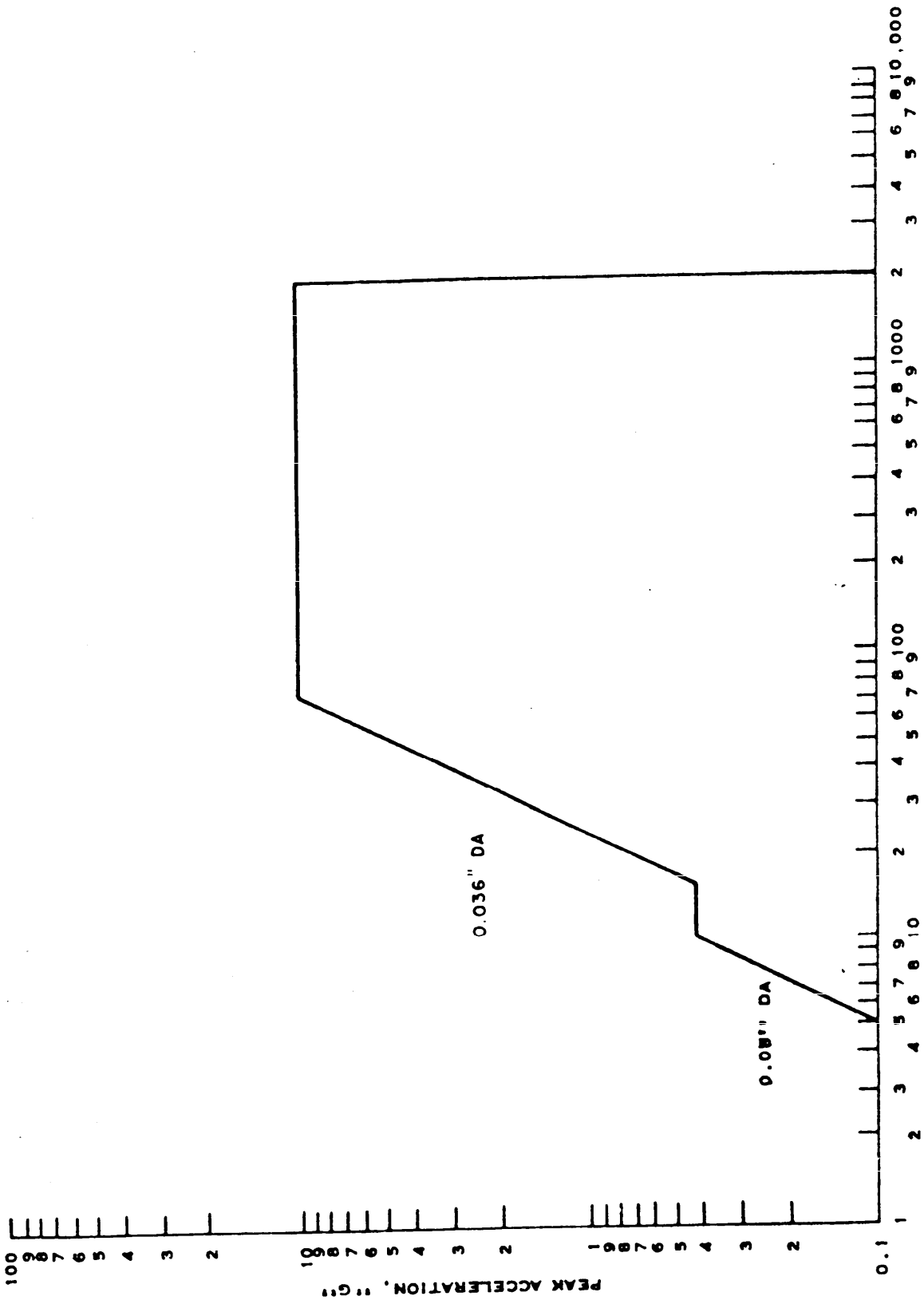


Figure 2. Vibration Curve, Sweep Generator.

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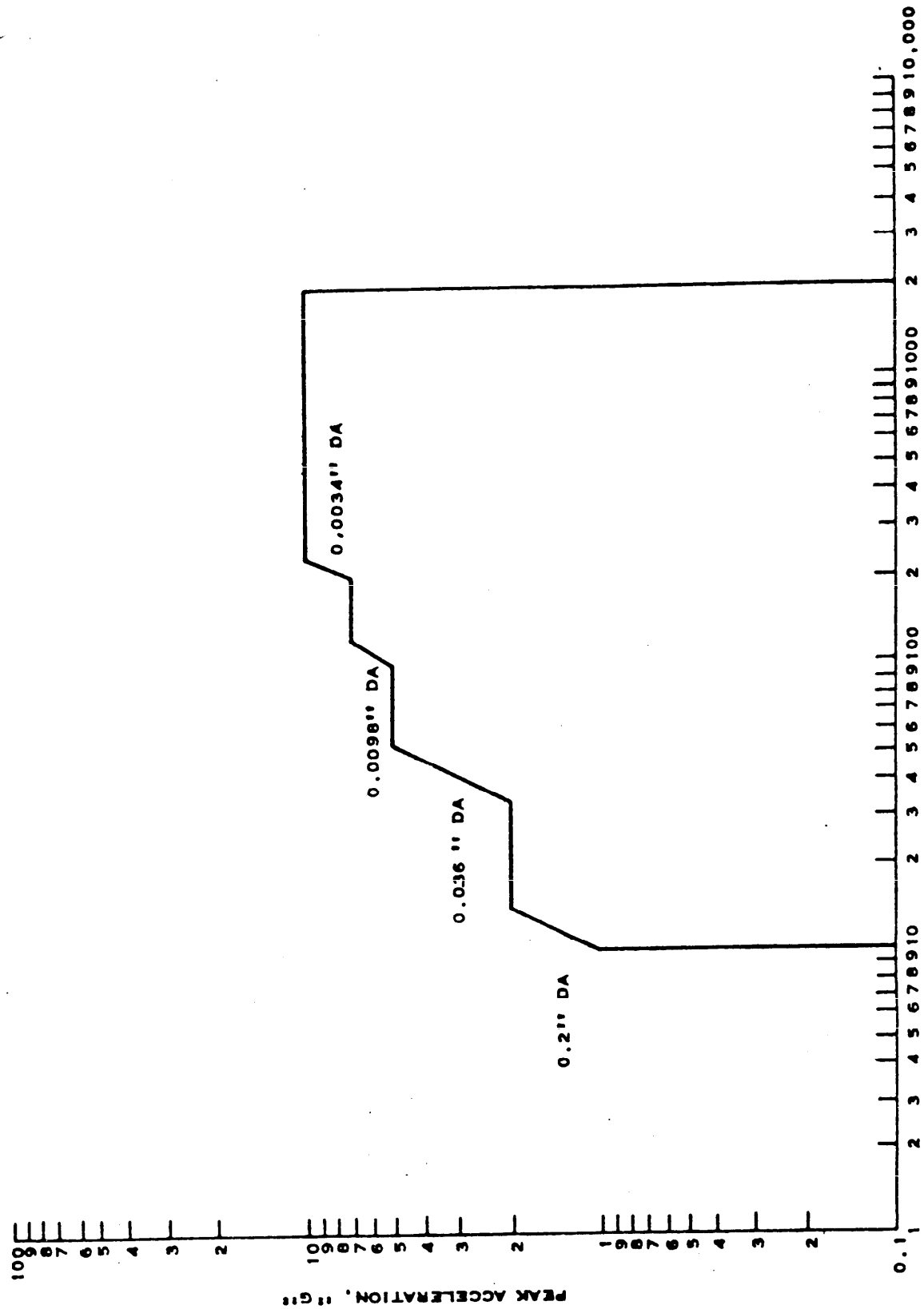


Figure 3. Vibration Curve, Fault Locator and Radar Set Controls

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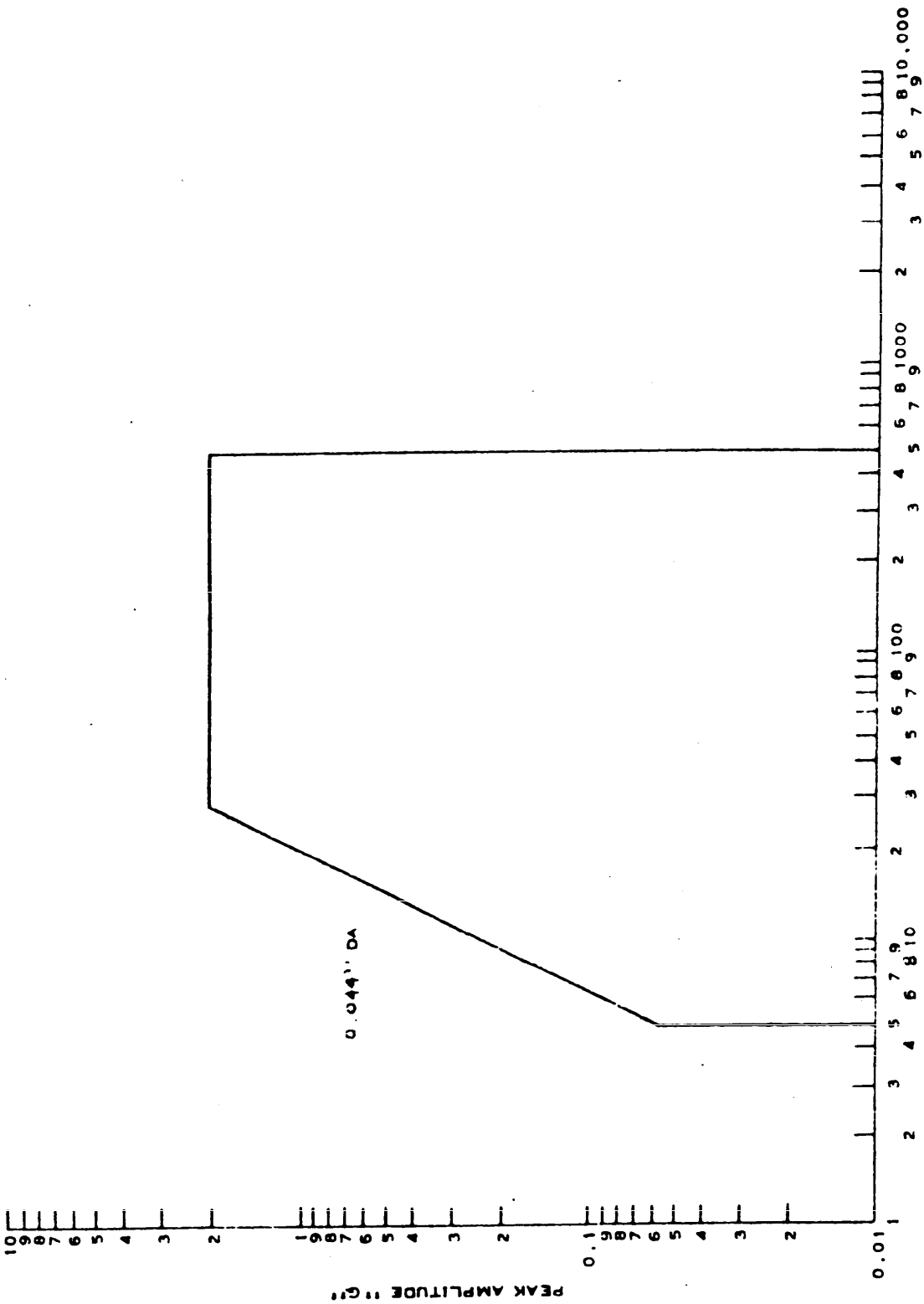


Figure 4. Vibration Curve, Indicator

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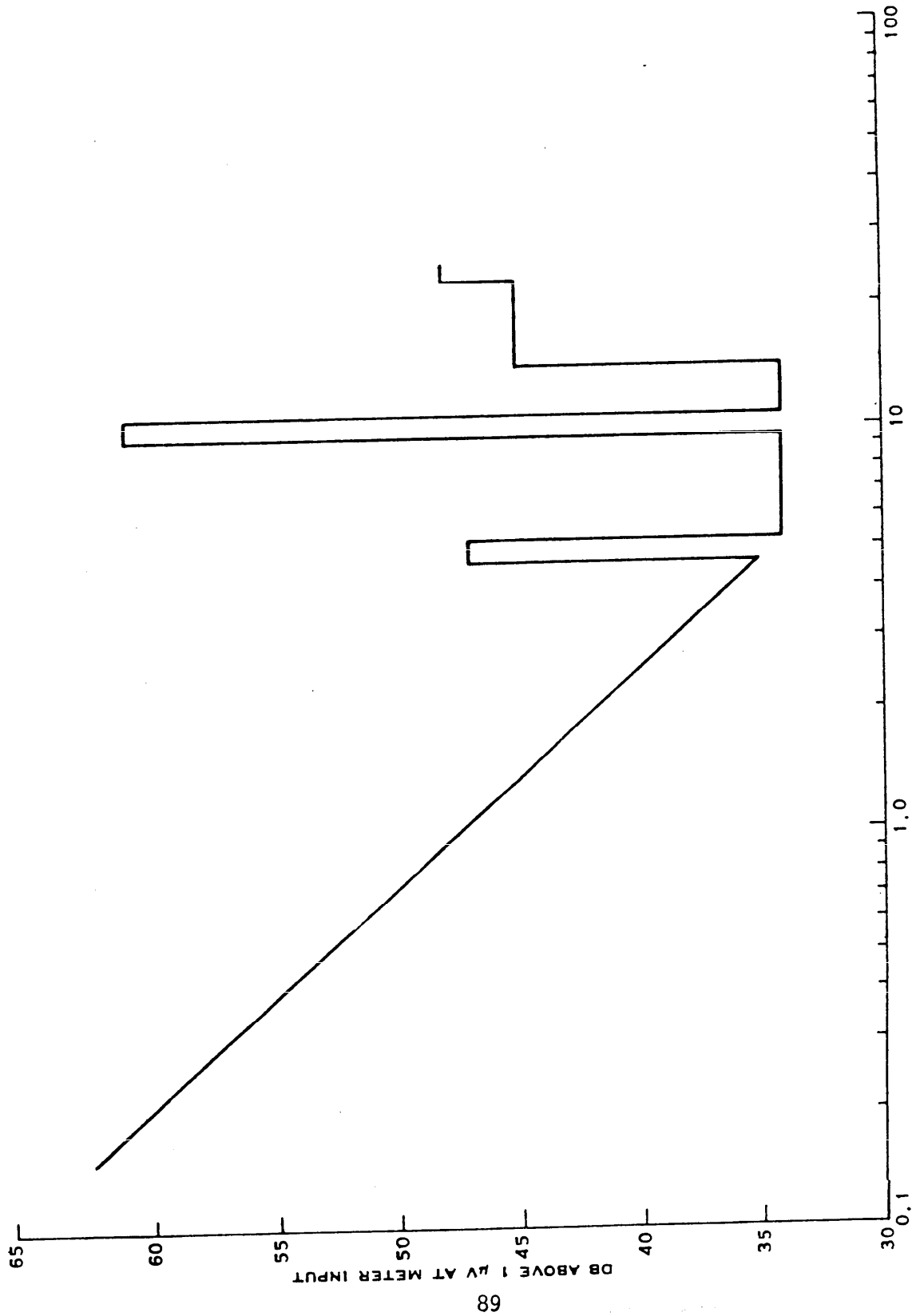


Figure 5. Narrow Band (CW) Conducted Interference Limits Using Stabilization Networks.

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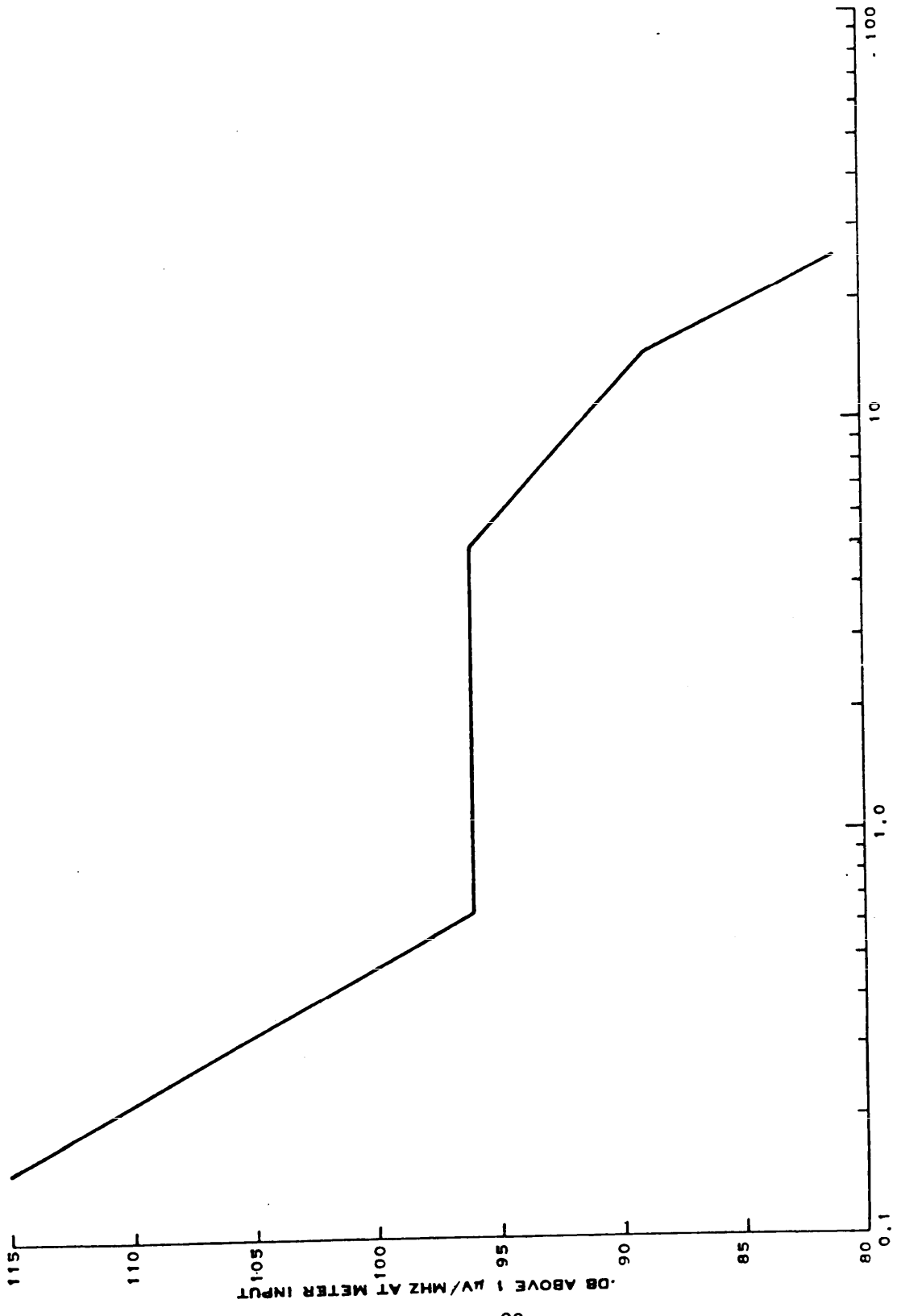


Figure 6. Broad Band and Pulsed CW Conducted Interference Limits Using Stabilization Networks.

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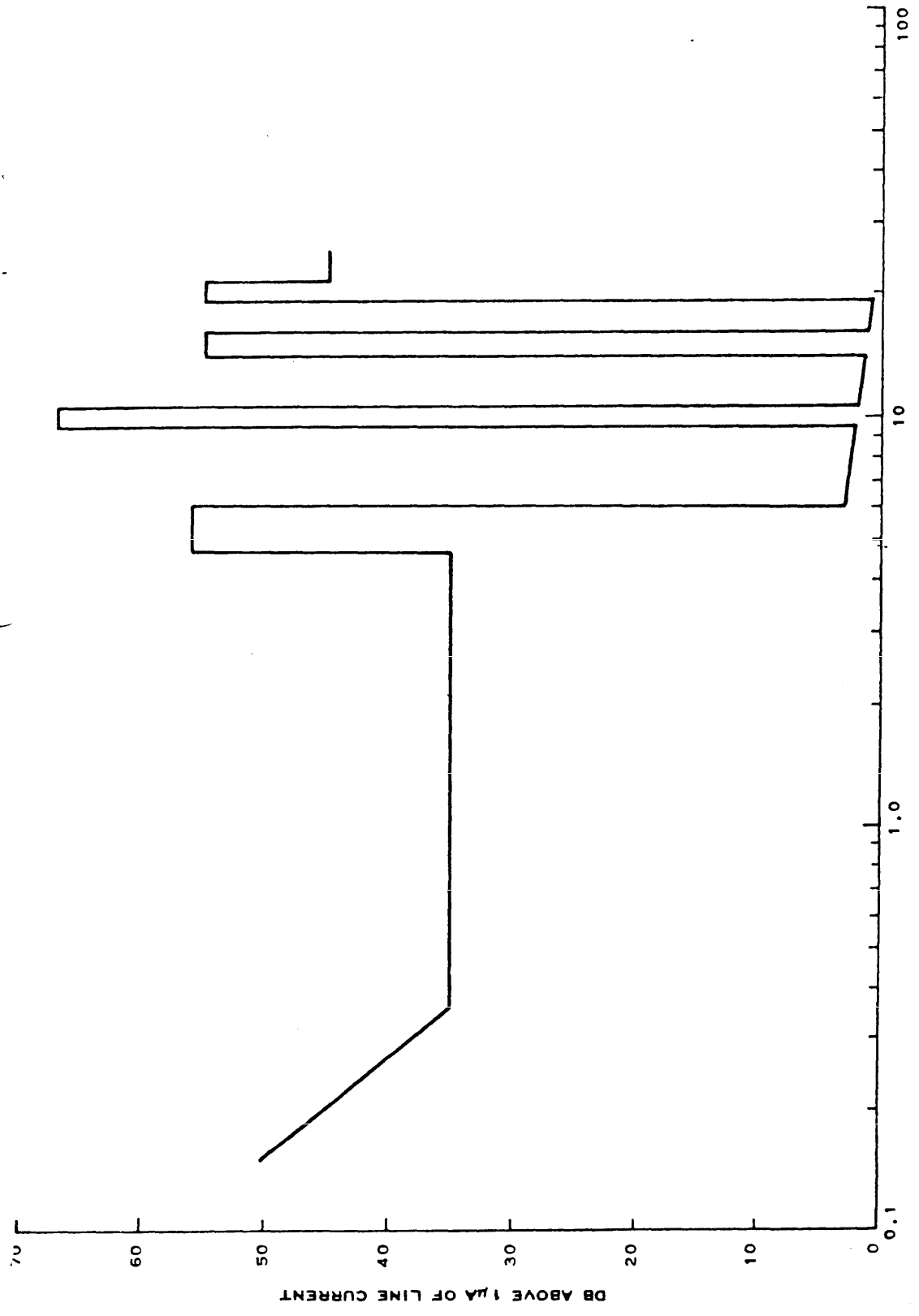


Figure 7. Narrow Band (CW) Conducted Interference Limits Using Current Probe

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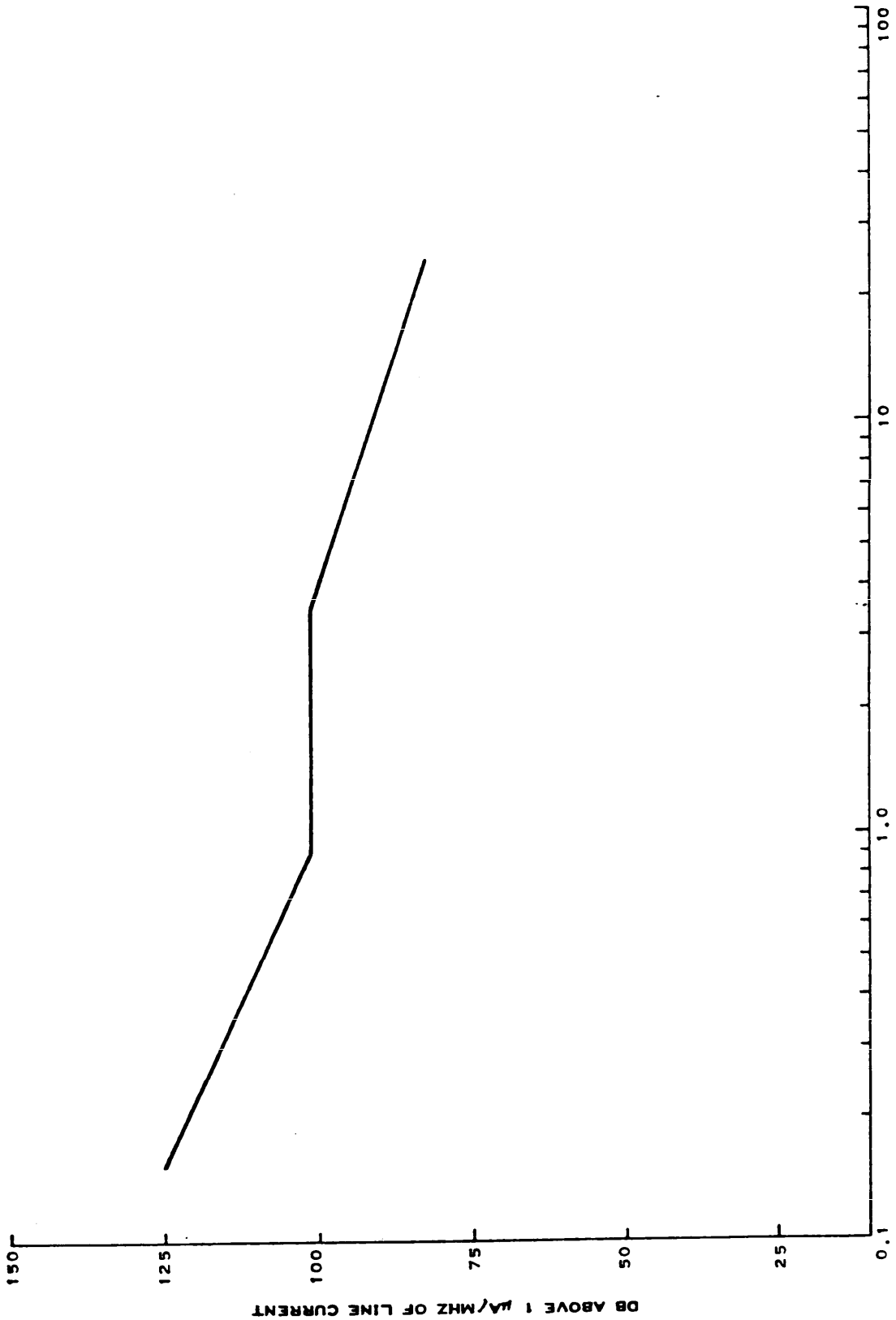


Figure 8. Broad Band and Pulsed CW Conducted Interference Limits Using Current Probe

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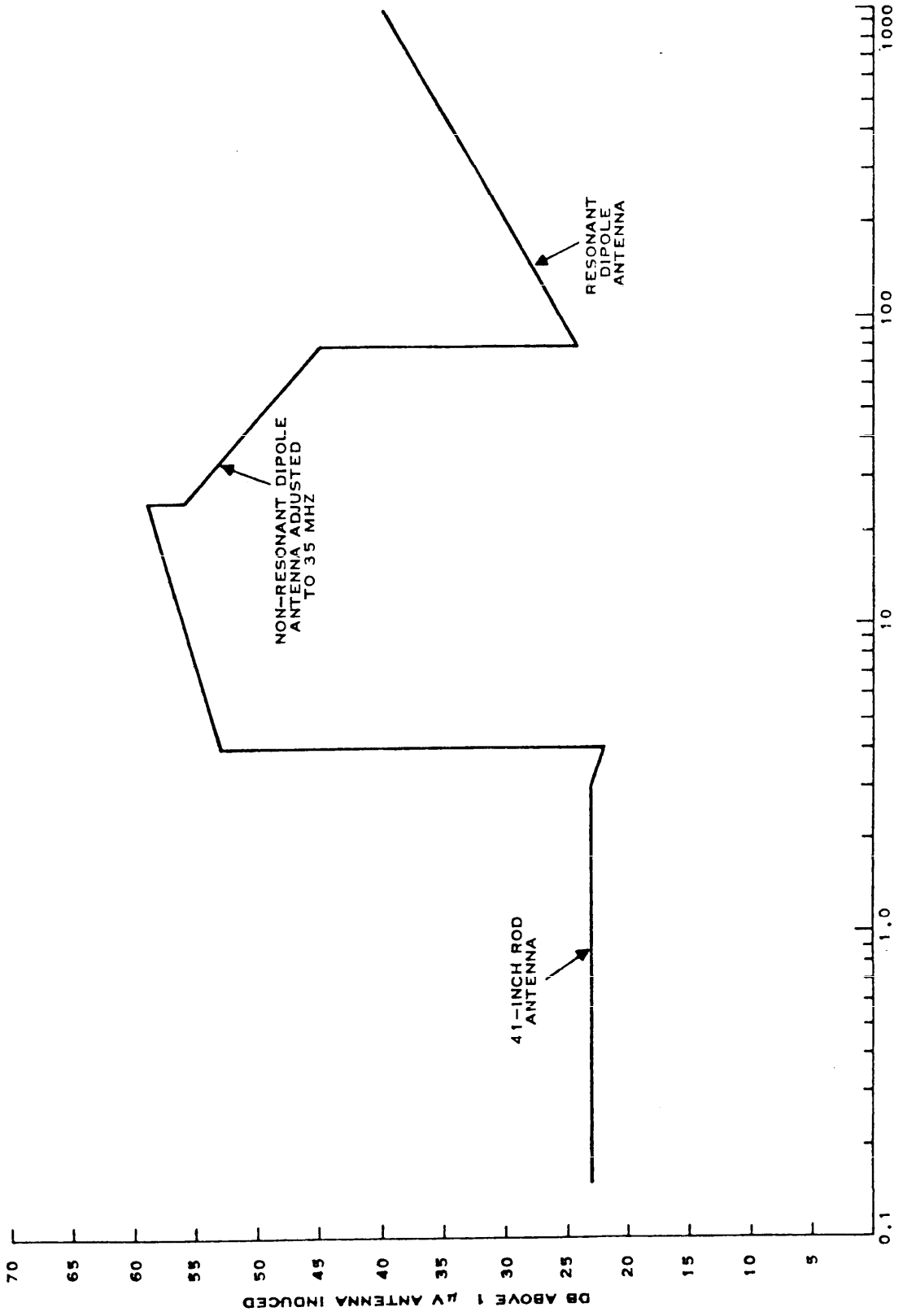


Figure 9. Narrow Band (CW) Radiated Interference Limits

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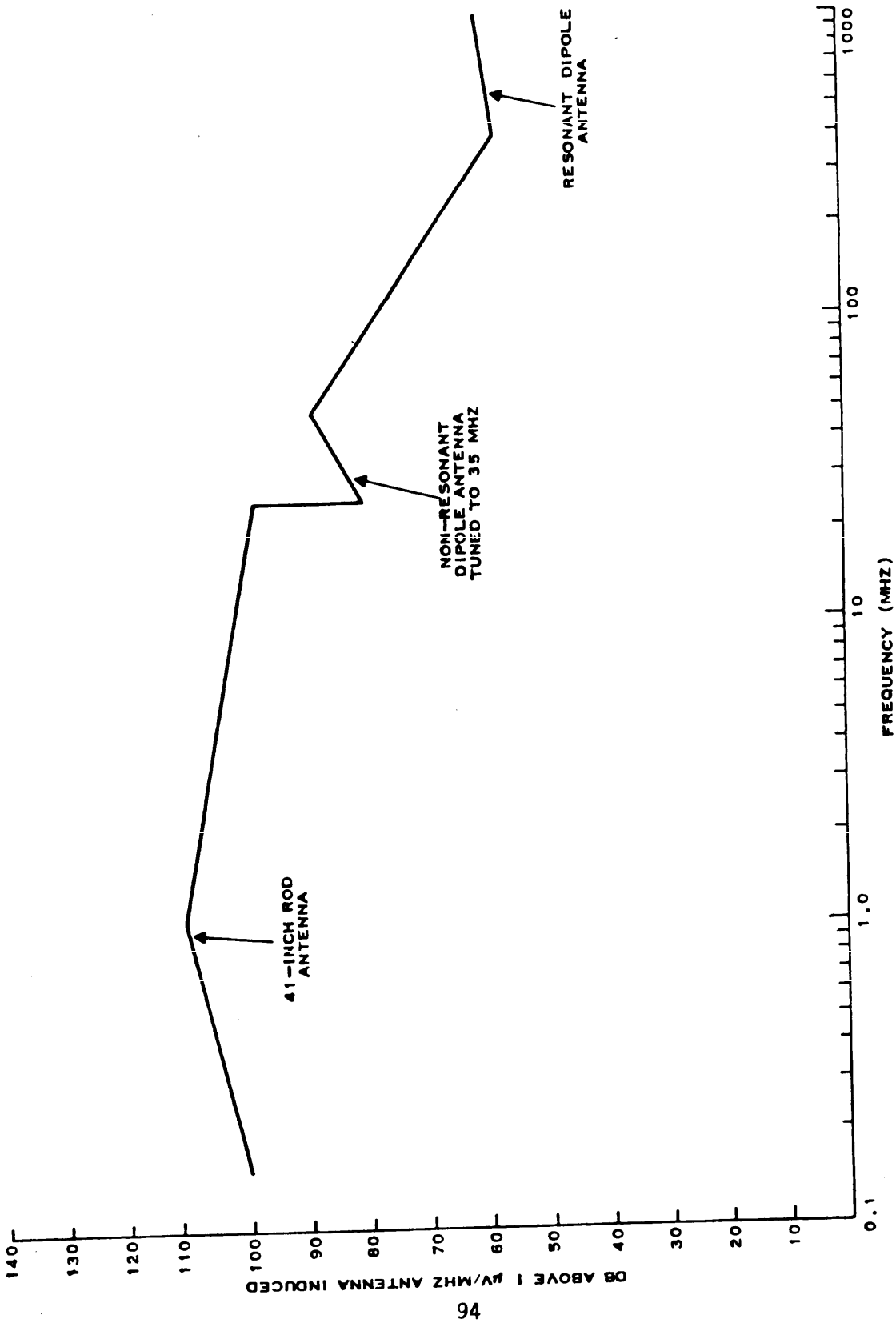


Figure 10. Broad Band and Pulsed CW Radiated Interference Limits.

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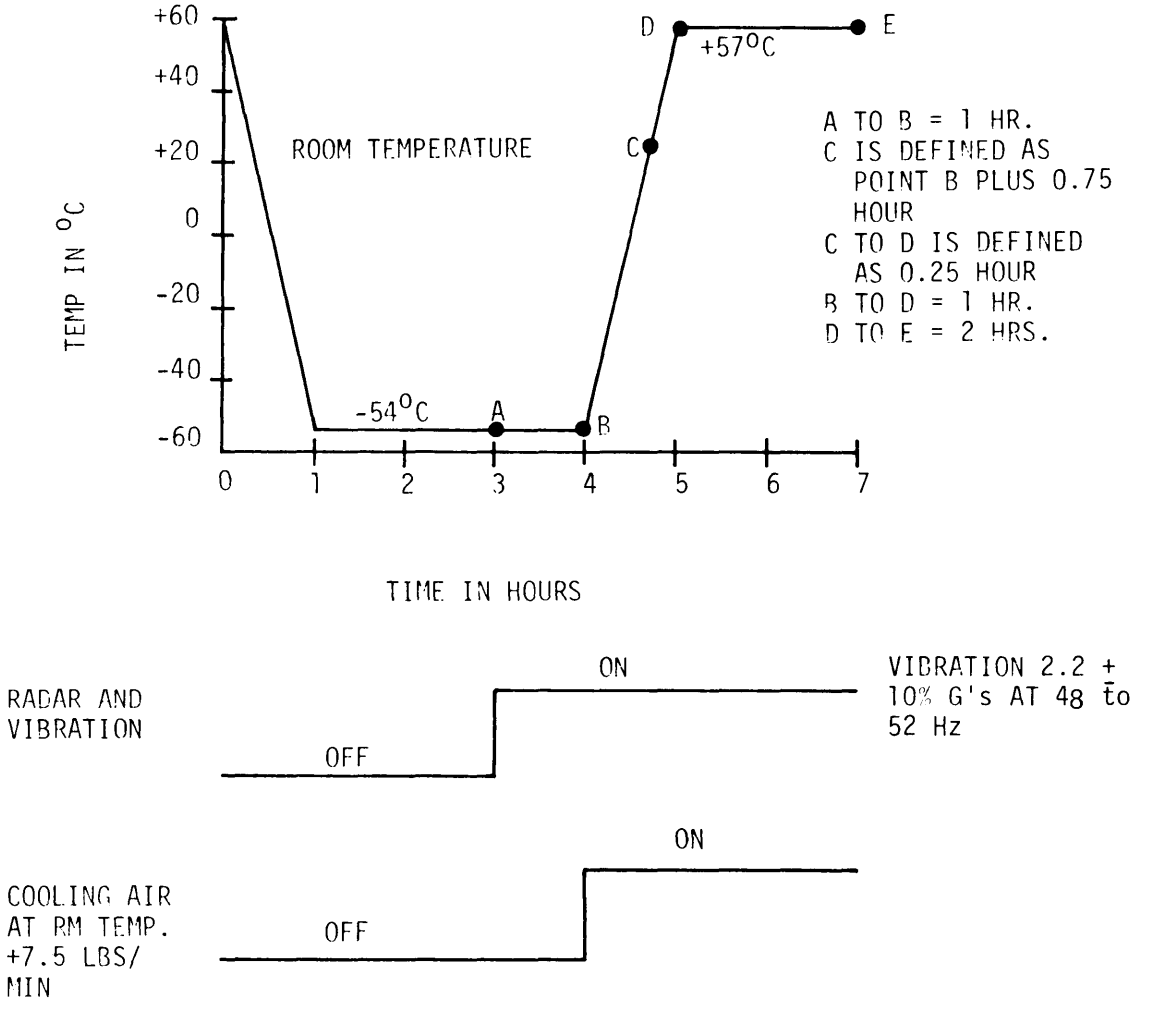
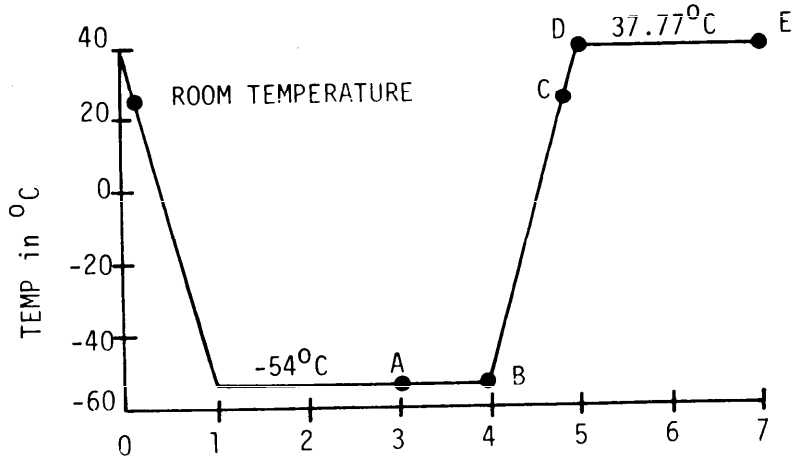


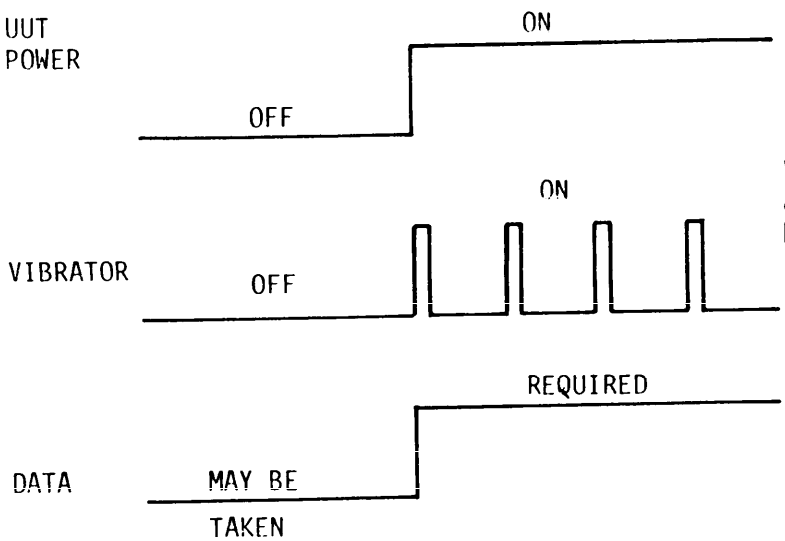
Figure 11. MANUFACTURING CONDITIONING THERMAL AND VIBRATION PROFILE SYSTEM BURN-IN.

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DESIGN ANALYSIS



A to B = 1 HR.
C is Defined as POINT B PLUS 0.86 HR.
C to D is Defined as 0.14 HOUR.
B to D = 1 HR
D to E = 2 HRS.



VIBRATION = 2.2 + 10% G's at 48 to 52 Hz FOR 10 MINUTES EACH ON PERIOD

Figure 12. Thermal and Vibration Profile Forward Assembly Group.

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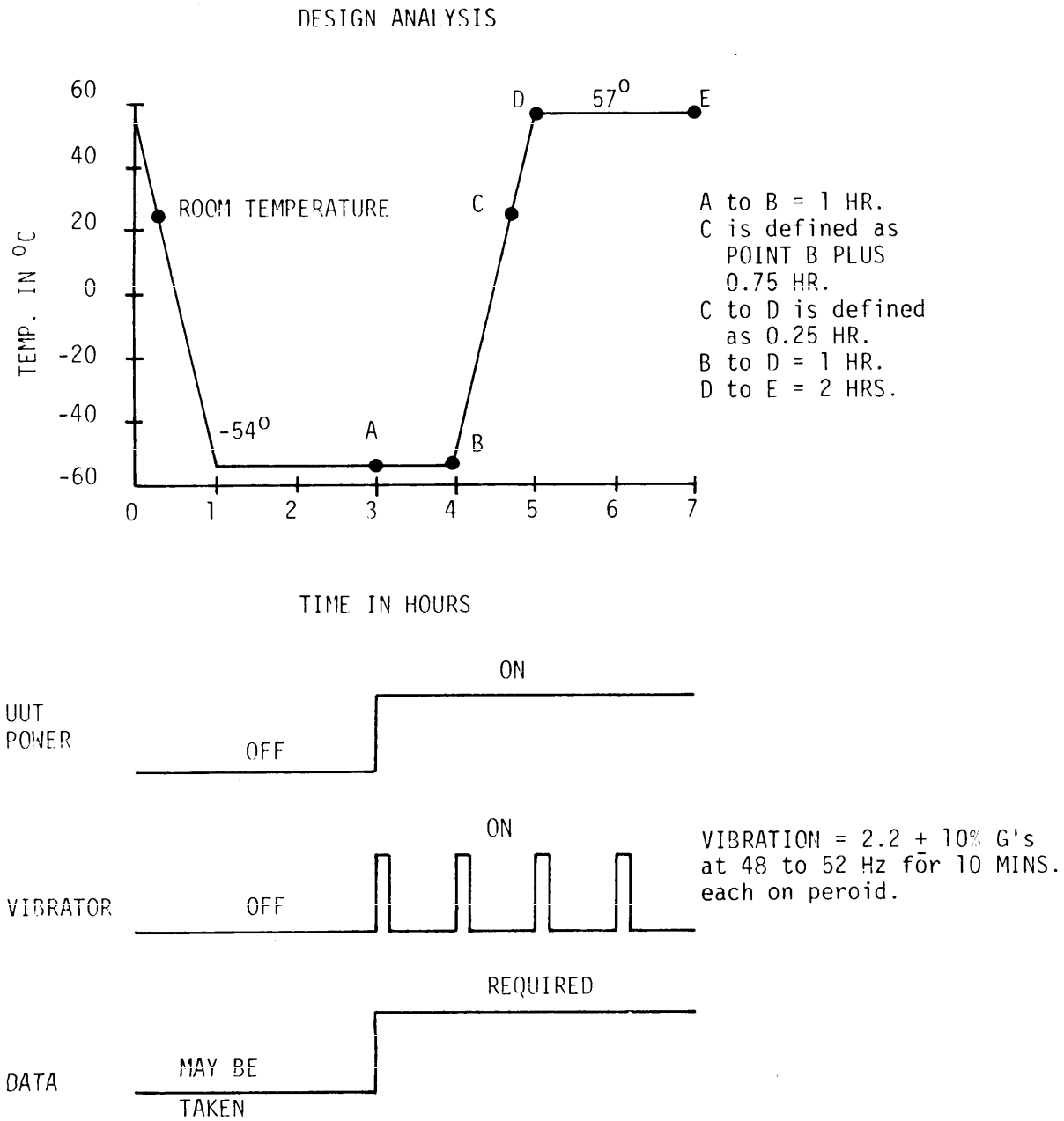


Figure 13. Thermal and Vibration Profile Cockpit Group.

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